Application of Operator Perspective in Enumeration of Multidimensional Random Walks

by

Aleksandar Vlasev

B.Sc. Mathematics, Simon Fraser University, 2012

A THESIS SUBMITTED IN PARTIAL FULFILLMENT
OF THE REQUIREMENTS FOR THE DEGREE OF

Master of Science

in

THE FACULTY OF SCIENCE
(Mathematics)

The University of British Columbia
(Vancouver)

April 2015

© Aleksandar Vlasev, 2015

Abstract

In this work we explore the the application of operators to enumeration of multidimensional random walks with a focus on walks which interact with the boundary hyperplanes. Using operators makes the derivation of the functional equations a lot easier and some preliminary work shows that using operators is the way to go when solving problems of this nature.

Up to 350 words

Preface

At University of British Columbia (UBC), a preface may be required. Be sure to check the Graduate and Postdoctoral Studies (GPS) guidelines as they may have specific content to be included.

Table of Contents

Abstract . . . . . **ii**

Preface . . . . . **iii**

Table of Contents . . . . . **iv**

List of Tables . . . . . **vii**

List of Figures . . . . . **viii**

Glossary . . . . . **ix**

Acknowledgments . . . . . **x**

0.1 Brief Note on Notation 1

1 Introduction to Methods - 1D random walks **3**

1.1 Derivation of the functional equation 3

1.2 Solving the functional equation for various s 8

1.2.1 The $s_{-1} = 0$ case. 9

1.2.2 The $s_{-1} \neq 0$ case. 10

1.2.3 1-dimensional case conclusion 14

2 Operators **15**

2.1 Definitions 15

2.2 Commutativity properties 16

2.3 Kernels, eigenvalues and eigenfunctions 18

2.3.1 Operator L . . . 18
2.3.2 Operator E . . . 18
2.3.3 Operator B . . . 19
2.3.4 Compounded operators . . . 19

3 Higher-dimensional labelled walks . . . 20
3.1 Using Operators to Derive the Functional Equation . . . 20
3.2 Generalizations to the functional equation . . . 22
3.3 Coefficient Extraction . . . 24
3.4 Further functional equations . . . 26
3.5 Using the formal inverse for iteration . . . 28

4 Specializing to 2 dimensions . . . 29
4.1 Introduction . . . 29
4.2 Investigating relationships between G, G_1, G_2 and $G_{1,2}$ . . . 30
4.3 Simple cases: Walks never leaving the origin and walks leaving the origin but never returning . . . 31
4.4 Simple case: Factoring S into a product . . . 33
4.5 Kernel Group Considerations . . . 36

5 Solving basic 2D cases . . . 39
5.1 Cases 1.1, 1.2, 1.3 . . . 39
5.2 Case 2.2 - group of order 1 . . . 40
5.3 Case 2.3 - group of order 2 . . . 42

6 Solving case 3.3 - supersymmetric models of order 4 . . . 44
6.1 Cartesian model: steps N, S, E, W . . . 45
6.2 Diagonal model: steps NE, NW, SE, SW . . . 46
6.3 The almost-full model: missing steps E and W . . . 54
6.4 The full model: all steps available . . . 54
6.5 Conjectured relationships depending on $S_{1,2}^{-1,-1}$ . . . 54

7 A little aside . . . 55

8 How To Use the template 58
8.1 Suggested Thesis Organization 59
8.2 Making Cross-References 59
8.3 Managing Bibliographies with BibTEX 60
8.3.1 Describing References 60
8.3.2 Citing References 61
8.3.3 Formatting Cited References 62
8.4 Typesetting Tables 62
8.5 Figures, Graphics, and Special Characters 63
8.6 Special Characters and Symbols 64
8.7 Changing Page Widths and Heights 65
8.7.1 The `geometry` Package 65
8.7.2 Changing Page Layout Values By Hand 66
8.7.3 Making Temporary Changes to Page Layout 66
8.8 Keeping Track of Versions with Revision Control 67
8.9 Recommended Packages 67
8.9.1 Typesetting 67
8.9.2 Figures, Tables, and Document Extracts 68
8.9.3 Bibliography Related Packages 68
8.10 Moving On 69

A Supporting Materials 70

List of Tables

Table 8.1 Available `cite` variants; the exact citation style depends on whether the bibliography style is numeric or author-year. . . . 62
Table 8.2 Useful LaTeX symbols 65

List of Figures

Figure 8.1 Proof of LaTeX's amazing abilities 64

Glossary

This glossary uses the handy `acroynym` package to automatically maintain the glossary. It uses the package's `printonlyused` option to include only those acronyms explicitly referenced in the LaTeX source.

CTAN The Common TeX Archive Network

DOI Document Object Identifier (see http://doi.org)

GPS Graduate and Postdoctoral Studies

RCS Revision control system, a software tool for tracking changes to a set of files

URL Unique Resource Locator, used to describe a means for obtaining some resource on the world wide web

Acknowledgments

I wish to say thank you to everyone who has helped me along my graduate school journey.

Thank you Victoria Zhu for supporting me throughout all these years. Without you love and support I would have quit many times! You were always there for me when I had doubts and wanted to give up.

Thank you Andrew Rechnitzer for taking me under your wing and believing in me. You are the man!

Thank you Kalle Karu. Without your support I wouldn't have made it this far. Your generousity will be forever appreciated!

Thank you mom and dad for believing in me, guiding me and supporting me in all the times of need. There's nothing quite like hearing you say that I can do it!

Thank you Tony Chong, Gloria Chien and Vlad Bohdanov for being my coaches and mentors, and helping me on the path to become a champion!

Also a big thank you to all my friends for letting me rant so many times!

Last but not least I want to thank UBC and the mathematics department especially for supporting me through thick and thin and somehow making things work for me whenever all hope was lost.

If I forgot anybody...you can just kill me.

0.1 Brief Note on Notation

We'll investigate the generating functions for quarter-plane walks where we track how many times these walks touch the walls.

- Q is the generating function.
- Powers of variables x_i keep track of the i-th coordinate of the endpoint of a random walk. For example the monomial $x_1^3 x_2^1 x_4^7$ counts how many walks end at $(3,1,0,7)$.
- Powers of variables a_i keep track of the number of times a random walk passes through a point with i-th coordinate equal to zero. For example the monomial $a_1^2 a_2$ counts how many times the random walks have passed through points with coordinates $(0,j)$ and $(i,0)$ respectively.
- Powers of t count the length of random walks.
- Variables with overline are reciprocals like $\overline{x_1} = 1/x_1$.
- S is the Laurent polynomial generating function for the allowed step set. For example for the steps $\{\text{N, E, SW}\}$ we have $S = x_2 + x_1 + \overline{x_1 x_2}$.
- We'll often use $b_i = a_1 - 1$ and $c_i = b_i/a_i = 1 - \overline{a_i}$.

Next we have some notation on extraction of coefficients

- $[d] = (1,\ldots,d)$ as a sequence.
- For a sequence $I = (i_1,\ldots,i_n)$ and a sequence $P = (p_1,\ldots,p_n)$, $x_I^P = x_{i_1}^{p_1}\ldots x_{i_n}^{p_n}$. For example $x_{[3]}^{(1,0,2)} = x_1^1 x_2^0 x_3^2 = x_1 x_3^2$. We'll usually use this notation for when we have general sequences, not in the case of the example which we would write out explicitly.
- $[u_I^P]$ stands for extracting the coefficient infront of u_I^P in the generating function expansion.
- $Q_I^J = [x_I^P]Q$ and similarly $S_I^J = [x_I^J]S$. For example if $I = (1,3)$, $J = 0,1$, (omiting parentheses) $Q_I^J = Q_{1,3}^{0,1}$. We will sometimes write $(Q_I^P)_J^R = Q_{I,J}^{P,R}$

- For a set I, $\mathbb{O}$ and $\mathbb{I}$ stand for sequences of $|I|$ zeros and ones respectively. For example for $I \subseteq [d]$ we might have $S_I^{-\mathbb{I}} Q_I^{\mathbb{O}}$. Analogously we will write $(Q_I^{\mathbb{I}})_J^{\mathbb{O}} = Q_{I,J}^{\mathbb{I},\mathbb{O}}$

So we have

$$Q(x_1, x_2, a_1, a_2; t) = \sum_{i,j,k} Q_{1,2}^{i,j} x_1^i x_2^j t^k$$

So far we won't extract a's because we have an unobvious choice between extracting a, b or c with various pros/cons. For example it may sometimes be easier to extract b_1^k than a_1^k but there's the added cost that the extracted term counts walks with varying powers of k. We usually won't extract with respect to t.

Chapter 1

Introduction to Methods - 1D random walks

1.1 Derivation of the functional equation

In the interest of working out higher dimensional labelled walks' problems it is very useful to first visit the 1-dimensional case which will illustrate a lot of the methods that will later come into play. The 1-dimensional case will also give us a clearer picture as to how things work in higher dimensions (although the picture becomes more complicated). We'll treat the problem using generating functions. We'll use lower case letters and not quite the full notation because it is not necessary. In one dimension, on one axis, we'll use steps of length at most 1. If we have a random walk ending at coordinate i we can either step back adding -1 to the coordinate, step forward adding $+1$ to the coordinate or staying in one place, adding 0 to the coordinate. This can be somewhat formalized by using polynomials. Ending a walk at coordinate i will be encoded with a monomial x^i. Then stepping backwards, forwards or staying in one place will be denoted with x^{i-1}, x^i and x^{i+1} respectively. This is the same as multiplying by $1/x = \bar{x}, 1, x$ in each case. We can encode this information is a Laurent step polynomial s defined as

$$s = s_{-1}\bar{x} + s_0 + s_1 x$$

Here s_i will be either 0 or 1 for any specific i, tracking whether we have allowed those kinds of steps or not. Keeping this in mind, there are 8 possible step sets because we have two options for each s_i. We keep track of the length of a random walk with powers of t and we keep track of its endpoint with powers of x. We'll also label every interaction with the origin by multiplying by a every time a walk steps on the origin. We'll keep the walks on the non-negative axis.

Long story short we can find the generating function in a very simple, brute force manner, even for higher dimensions. We start with the empty walk denoted by 1 since it starts and ends on the origin without taking a step, so no t, x or label a. Then try out each step from s and eliminate those that take us past the origin in the negative direction. If we step on the origin we multiply by a. Those walks that remain are then multiplied by t since they are not of length 1. We can proceed iteratively k tiems to obtain random walks on the non-negative axis of length k. Adding up the results allows us to write the generating function up to $O(t^k)$. It will prove very useful to think about operators so we can formalize this notion in terms of operators. For example, here we are taking walks of length $k-1$, trying all options of s (this is like multiplying by s), removing all the walks that end up on the negative axis (this is like removing the $\bar{x}$ term) and labeling those that end on the origin (this is like multiplying the x^0 coefficient by a). Let D stand for the operator that cleans up the walks and labels the valid ones. Then if q_k stands for the labelled walks of length k, then the k-th term of q is $q_k t^k$ and we have

$$q_{k+1}t^{k+1} = tD(sq_k t^k)$$

Thus we can write the generating function q iteratively

$$q = \sum_{k=0}^{\infty} t^k (Ds)^k (1)$$

where we're iteratively applying Ds to 1 to obtain all labelled walks. We can treat this operator Ds as a formal variable and write

$$q = \sum_{k=0}^{\infty} (tDs)^k (1) = \frac{1}{1-tDs}(1)$$

and then this gives us a functional equation for q

$$(1 - tDs)q = 1$$

There is another way of deriving this equation while not treating Ds as a formal variable. Instead of applying the operator iteratively we can apply it on the generating function as a whole. The product tsq is the generating for all random walks that end with x-coordinate greater or equal to -1 where some steps end up on x-coordinate 0 without being labelled. We can apply D on this product to remove the terms ending at x-coordinate -1 and label those that end up on x-coordinate 0. This leaves us with all the valid walks encoded in the generating function q without the empty walk 1. Therefore we can write a functional equation for q as

$$q = 1 + D(tsq) = 1 + tDs(q)$$

and we ca re-write this as $(1 - tDs)q = 1$ as before. Now, we have to say a few words about what the operator D actually is. First we are removing all the walks that end on x-coordinate equal to -1 which can be handled by the operator $E = 1 - \overline{x}[\overline{x}]$ which removes those walks. Then the labelling can be handled by the operator $L = 1 + (a-1)[x^0] = 1 + b[x^0]$ where $b = a - 1$. Therefore we can write $D = LE$

$$\begin{aligned} D = LE &= (1 - \overline{x}[\overline{x}])(1 + b[x^0]) \\ &= 1 + b[x^0] - \overline{x}[\overline{x}] - \overline{x}[\overline{x}]b[x^0] \\ &= 1 + b[x^0] - \overline{x}[\overline{x}] \end{aligned}$$

since the last term vanishes on all generating functions. We could have also applied the operators in reverse to get the same result. Thiscommutativity makes sense because on the level of the function they are operator interacting with different

monomials. We can now expand the functional equation to the following

$$\begin{aligned}1 &= q - tLE(sq) = q - t(sq + b(sq)_0 - \overline{x}(sq)_{-1}) \\ &= q - t(sq + b(s_{-1}q_1 + s_0q_0 + s_1q_{-1}) - \overline{x}(s_{-1}q_0)) \\ &= q - t(sq + b(s_{-1}q_1 + s_0q_0) - \overline{x}(s_{-1}q_0))\end{aligned}$$

since $q_{-1} = 0$. We can rewrite this equation in the following form

$$(1 - ts)q - t(bs_0 - \overline{x}s_{-1})q_0 - bts_{-1}q_1 = 1$$

Here $q_0 = q(0)$ and $q_1 = q'(0)$ where $q = q(x)$. In the interest of simplifying this equation we can eliminate the q_1 term by extracting the x^0 coefficient from the whole equation

$$\begin{aligned}1 &= q_0 - t(s_{-1}q_1 + s_0q_0 + s_1q_{-1}) - tbs_0q_0 - bts_{-1}q_1 \\ &= q_0 - t(s_{-1}q_1 + s_0q_0) - tbs_0q_0 - bts_{-1}q_1 \\ &= (1 - ts_0 - tbs_0)q_0 - (ts_{-1} + bts_{-1})q_1 \\ &= (1 - ats_0)q_0 - t(s_{-1} + bs_{-1})q_1 \\ &= (1 - ats_0)q_0 - ats_{-1}q_1\end{aligned}$$

We can solve this equation for q_1

$$ts_{-1}q_1 = \overline{a}((1 - ats_0)q_0 - 1)$$

and substitute the result into the functional equation (for now we don't worry about the possibility of $s_{-1} = 0$ because it leads to the same thing anyway).

$$\begin{aligned}1 &= (1 - ts)q - t(bs_0 - \overline{x}s_{-1})q_0 - c((1 - ats_0)q_0 - 1) \\ &= (1 - ts)q - (c(1 - ats_0) + bts_0 - t\overline{x}s_{-1})q_0 + c \\ &= (1 - ts)q - (c - bts_0 + bts_0 - t\overline{x}s_{-1})q_0 + c \\ &= (1 - ts)q - (c - t\overline{x}s_{-1})q_0 + c\end{aligned}$$

and therefore we can rewrite the equation as

$$(1-ts)q-(c-t\bar{x}s_{-1})q_0=\bar{a}$$

since $1-c=1-(1-\bar{a})=\bar{a}$. This process is rather complicated, even though it can be done for higher dimensional problems as well (as you can see in various previous approaches). Also there is an issue that we need to consider every time we are solving an equation in order to substitute quantities back into the equation. This is where the operator notation comes to the resque. It turns out the the L operator is invertible with inverse given by

$$L^{-1}=M=1-c[x^0]$$

and we can apply it on the functional equation. The RHS is simply $M(1)=1-c=\bar{a}$ and the LHS is

$$\begin{aligned}M(1-tLEs)q&=Mq-tEsq\\&=(q-cq_0)-t(sq-\bar{x}s_{-1}q_0)\\&=(1-ts)q-(c-t\bar{x}s_{-1})q_0\end{aligned}$$

and we've derived the same equation but in a much neater way. This will prove crucial later on when we derive a functional equation for higher dimensional problems. Although it can be done directly with a lot of effort it's much better to just use the operator notation. The derivation is brought down from about a dozen pages to one or two. The operator notation will also allow us to solve some multidimensional problems directly or with the help of lower dimensional problems.

We can extract the x^0 coefficient from the final functional equation again and although we can do this directly, we'll use the operator notation to tidy up the process. This will show up later when dealing with higher dimensional problems

$$\begin{aligned}[x^0]M&=[x^0]-[x^0]c[x^0]=[x^0]-c[x^0]=\bar{a}[x^0]\\[x^0]E&=[x^0]-[x^0]\bar{x}[\bar{x}]=[x^0]\end{aligned}$$

Then we extract the x^0 from the functional equation

$$[x^0]\overline{a} = \overline{a} = [x^0](M - tEs)q = ([x^0]M - t[x^0]Es)q = (\overline{a}[x^0] - t[x^0]s)q = [x^0](\overline{a} - ts)q$$

Therefore the extraction leads to

$$[x^0](1 - ast)q = 1$$

We could extract higher coefficients from the functional equation too. Let $k \geq 1$. Then

$$[x^k]M = [x^k] - [x^k]c[x^0] = [x^k]$$
$$[x^k]E = [x^k] - [x^k]\overline{x}[\overline{x}] = [x^k]$$

Therefore

$$[x^k]\overline{a} = 0 = [x^k](M - tEs)q = ([x^k]M - t[x^k]Es)q = ([x^k] - t[x^k]s)q = [x^k](1 - ts)q$$

That is

$$[x^k](1 - ts)q = 0$$

and we can readily extract whichever coefficients we want. The first equation leads to

$$(1 - ats_0)q_0 - ats_{-1}q_1 = 1$$

and the second leads to

$$q_k - t(s_{-1}q_{k+1} + s_0q_k + s_1q_{k-1}) = 0$$
$$(1 - s_0t)q_k - t(s_{-1}q_{k+1} + s_1q_{k-1} = 0$$

1.2 Solving the functional equation for various s

In this section we'll solve all 8 one-dimensional problems fully. Some cases are easier to deal with than others. As we'll see later in the higher dimensional cases, some problems are easierto solve than others. This difficulty tends to scale up with

the inclusion of certain steps with negative direction.

1.2.1 The $s_{-1} = 0$ case.

This condition allows us to solve explicitly for q_0.

$$q_0 = \frac{1}{1 - as_0t}$$

We can substitute this expression in the functional equation and obtain

$$q = \frac{\overline{a}}{1 - st}\left(1 + \frac{b}{1 - as_0t}\right) = \frac{1 - s_0t}{(1 - st)(1 - as_0t)}$$

This takes care of half of the cases! What happens when we set $a = 1$, corresponding to setting $b = 0$ or extracting the b^0 coefficient from the equation? This corresponds to unlabelling the random walks. We obtain

$$[b^0]q = \widehat{q} = \frac{1 - s_0t}{(1 - st)(1 - s_0t)}$$

We'll note that

$$(1 - as_0t)q - (1 - s_0t)\widehat{q} = 0$$

This can also be written in operator form with $B = 1 - [b^0]$

$$B((1 - as_0t)q) = 0$$

This relates the labelled generating function to the unlabelled generating function! it says we can obtain the labelled generating function from the unlabelled one by the following relation

$$q = \frac{1 - as_0t}{1 - s_0t}\widehat{q}$$

We can also establish such as relationship for q_0 although the derivation here will be trivial. Extracting $[x^0]$ from the equation leads to

$$B((1 - as_0t)q_0) = 0$$

and

$$q_0 = \frac{1 - as_0 t}{1 - s_0 t} \widehat{q_0}$$

The relationship between q and $\widehat{q}$ is no accident but it tends to happen fairly often in the higher dimensional cases that we've considered. The functional equation in the $s_{-1} = 0$ case is easier to solve most likely because when $s_{-1} = 0$, walks don't go beyond the origin in the negative direction. Consequently the E operator acts as the identity on the generating function. and the functional equation reads as

$$(M - ts)q = q - cq_0 - tsq = (1 - ts)q - cq_0 = \overline{a}$$

1.2.2 The $s_{-1} \neq 0$ case.

The functional equation in the $s_{-1} \neq 0$ is also not difficult to solve. We'll try to apply the kernel method whenever possible. It requires finding an x value that makes the kernel $1 - ts$ equal to zero in the functional equation

$$(1 - ts)q - (c - t\overline{x})q_0 = \overline{a}$$

Such a solution will exist since $s_{-1} = 1$ and therefore the equation $1 - ts = 0$ is non-trivial. Let one such solution be $x = \sigma$ and suppose that we can plug it into the functional equation. Then the kernel vanishes, taking q away with it and we obtain

$$-(c - t\overline{\sigma})q_0 = \overline{a}$$

We can solve for q_0 explicitly again

$$q_0 = \frac{\overline{a}}{\overline{\sigma}t - c}$$

We can now solve for q

$$q = \frac{\overline{a}}{1 - ts}\left(1 - \frac{t\overline{x} - c}{t\overline{\sigma} - c}\right) = \frac{t}{1 - ts}\left(\frac{\overline{x} - \overline{\sigma}}{b - a\overline{\sigma}t}\right)$$

We note that we can apply the B operator again

$$B((a\overline{\sigma}t-b)q)=0$$

or written out explicitly

$$(a\overline{\sigma}t-b)q-(\overline{\sigma}t)\widehat{q}=0$$

This also allows us to write the labelled generating function in terms of the unlabelled generating function

$$q=\frac{\overline{\sigma}t}{a\overline{\sigma}t-b}\widehat{q}=\frac{t}{at-b\sigma}\widehat{q}$$

We can also work with q_0 at this stage. We find

$$[x_0]B((a\overline{\sigma}t-b)q)=B((a\overline{\sigma}t-b)q_0)=0$$

too. This ends up being the same as

$$(a\overline{\sigma}t-b)q_0-(\overline{\sigma}t)\widehat{q_0}=0$$

with solution

$$q_0=\frac{t}{at-b\sigma}\widehat{q_0}$$

The final task for this section will be to find what σ will be for various cases. We already know $s_{-1}=0$ so the result depend on s_0 and s_1. In general the kernel is zero when

$$\begin{aligned} ts &= 1 \\ t(s_{-1}+s_0x+s_1x^2) &= x \\ ts_1x^2-(1-s_0t)x+ts_{-1} &= 0 \end{aligned}$$

When $s_1\neq 0$ this is quadratic with two roots that are functions of t. We have to pick the one which has no singularity at $t=0$. When $s_1=0$ we have a linear equation which may have a single root. When $s_1=0$ we have the equation $(1-s_0t)x=t$ so

the single root is

$$\sigma = \frac{t}{1 - s_0 t}$$

When $s_1 = 1$ we have the equation $tx^2 - (1 - s_0 t)x + t = 0$ with roots

$$\sigma = \frac{1 - s_0 t - \sqrt{(1 - s_0 t)^2 - 4t^2}}{2t}, \quad \sigma^* = \frac{1 - s_0 t + \sqrt{(1 - s_0 t)^2 - 4t^2}}{2t}$$

The two roots multiply to 1 so $\overline{\sigma} = \sigma^*$. Later we shall have a little table tabulating all the results. Note that in the relationship between the labelled and unlabelled q_0 we have

$$q_0 = \frac{t}{at - b\sigma} \widehat{q_0}$$

where σ is a function containing a square root, i.e. not a polynomial. However we can go at the problem from a different angle to obtain a polynomial $f(a,t)$ for which $B(fq_0) = c$ which is not quite zero but it's a simpler relationship. This will also illustrate the group approach together with the kernel method. The two cases that we need to work on both have $s_{-1} = s_1 = 1$ with s_0 being 0 or 1. This means that s is left invariant under sending $* : x \mapsto \bar{x}$. We can apply this on the functional equation and now have a system of two functional equations

$$(1 - ts)q - (c - t\bar{x})q_0 - \bar{a} = 0$$
$$(1 - ts)q^* - (c - tx)q_0 - \bar{a} = 0$$

We can multiply the first by $c - tx$, the second by $c - t\bar{x}$ and subtract the equations to eliminate the q_0 term. We can divide everything out by $1 - ts$ and finally obtain

$$(c - tx)q(x) - (c - t\bar{x})q(\bar{x}) - \frac{(c - tx - c + t\bar{x})}{1 - ts}\bar{a} = 0$$
$$(c - tx)q(x) - (c - t\bar{x})q(\bar{x}) - \bar{a}\delta = 0$$

where

$$\delta = \frac{t(\bar{x} - x)}{1 - ts}$$

and crucially it doesn't depent on a. We extract the x^1 coefficient from the equation to obtain an equation relating q_1 and q_0

$$cq_1 - tq_0 - \overline{a}\delta_1 = 0$$

We can set $a = 1$ here so we get

$$-\delta_1 = t\widehat{q_0}$$

and we can bring this back to the original equation to get

$$\begin{aligned} cq_1 &= tq_0 - \overline{a}t\widehat{q_0} \\ bq_1 &= t(aq_0 - \widehat{q_0}) \end{aligned}$$

Earlier we had the equation

$$(1 - as_0t)q_0 = 1 + as_{-1}tq_1 = 1 + atq_1$$

which we can solve for q_1 this time and insert the solution into the above equation

$$b((1 - as_0t)q_0 - 1) = t^2a(aq_0 - \widehat{q_0})$$

Collecting coefficients and simplifying leads us to

$$\begin{aligned} (b(1 - as_0t) - a^2t^2)q_0 - (-at^2)\widehat{q_0} &= b \\ (c(1 - as_0t) - at^2)q_0 - (-t^2)\widehat{q_0} &= c \end{aligned}$$

which gives $f(a,t) = c(1 - as_0t) - at^2$ so that

$$B((c(1 - as_0t) - at^2)q_0) = c.$$

With two different approaches we obtained slightly different relationships upon applying the operator B.

1.2.3 1-dimensional case conclusion

In this section we were able to find explicit solutions for q_0, q and also B-operator relationships for q and $\widehat{q}$. In higher dimensions this may not be quite possible but we should still be able to find B-operator relationships for the equivalent of q_0 and $\widehat{q_0}$. Here is a table showing the results of this section. Here is a table showing detailed results for each of the 8 possible cases of random walks restricted to the non-negative axis.

Chapter 2

Operators

In this section we'll make our way towards the larger problem of higher-dimensional labelled walks. The methods will be similar to those in the section on the one-dimensional problem but their complexity will rise substantially. Without the operator notation even the derivation of the functional equation is quite laborious. It's probably worth mentioning that we found the derivation the hard way first before we discovered we can use operators to simplify the process. What once took about a dozen pages now takes a just few. In this section we'll explore the properties of the operators that interest us, showing when the commute, if they have inverses, what their eigenvalues and eigenvectors are and so on.

2.1 Definitions

We'll work with many dimensions directly as that doesn't really change the situation very much. Let d be the number of dimensions. A walk in d dimensions is valid with respect to coordinate x_i if it never steps on a point with negative x_i coordinate. A walk in d dimensions is valid if it is valid with respect to all x_i coordinates, that is it remains in the non-negative orthant. Let

$$E_i = 1 - \overline{x}_i[\overline{x}_i]$$

be the extraction operator with respect to x_i. Applying it on an expression like SQ yields the generating function for x_i-valid walks. Similarly let

$$L_i = 1 + b_i[x_i^0]$$

be the labelling operator with respect to the $x_i = 0$ hyperplane. This operator labels walks that end on the $x_i = 0$ hyperplane with the label a_i. Let

$$M_i = 1 - c_i[x_i^0].$$

We can call this operator the de-labelling operator because it is the inverse of L_i, which we'll show later. Let

$$B_i = 1 - [b_i^0].$$

We don't have a name for this operator yet but we use it when trying to establish a relationship between labelled and unlabelled quantities. Maybe we can call it the relator? Next let

$$P = P_d \dots P_1$$

For any operator P, omitting the subscript will mean

$$P = P_d \dots P_1$$

2.2 Commutativity properties

Firstly we have that if $i \neq 0$

$$[x^i][x^k] = 0$$

Then

$$[x^i]x^j[x^k] = \delta_{i,j}[x^k]$$

This helps establish the commutativity properties of all the operators we have so far E_i, L_i, M_i, D_i, B_i. In general, if we have operators A and B given by

$$A(x) = \sum_k A_k x^k [x^k], \quad B(x) = \sum_k B_k x^k [x^k]$$

then

$$\begin{aligned}
A(x)B(x) &= \left(\sum_k A_k x^k [x^k]\right)\left(\sum_k B_k x^k [x^k]\right) \\
&= \sum_{i,j} A_i x^i [x^i] B_j x^j [x^j] \\
&= \sum_{i,j} A_i B_j x^i \delta_{i,j} [x^j] \\
&= \sum_k A_k B_k x^k [x^k]
\end{aligned}$$

We obtain the same result if we reverse A and B, so they commute. Now

$$\begin{aligned}
A(x)B(y) &= \left(\sum_k A_k x^k [x^k]\right)\left(\sum_k B_k y^k [y^k]\right) \\
&= \sum_{i,j} A_i x^i [x^i] B_j y^j [y^j] \\
&= \sum_{i,j} A_i B_j x^i y^j [x^i y^j] \\
&= \sum_{i,j} B_j y^j [y^j] A_i x^i [x^i] \\
&= \left(\sum_k B_k y^k [y^k]\right)\left(\sum_k A_k x^k [x^k]\right) \\
&= B(y)A(x)
\end{aligned}$$

So they commute. Our operators are of the form $1 + A(x)$ so they will commute. In general, as long as each coefficient extraction is only doing things on the level of the power of x they are extracting, the operators will commute. If there is some mixing we'll potentially not have commuting operators. This allows us to later create easy generalizations of the labelling problem so far.

2.3 Kernels, eigenvalues and eigenfunctions

2.3.1 Operator L

Operator L is invertible with inverse M

$$\begin{aligned} LM &= (1+b[x^0])(1-c[x^0]) \\ &= 1+b[x^0]-c[x^0]-b[x^0]c[x^0] \\ &= 1+(b-c-bc)[x^0] \\ &= 1+(a-1-1+\overline{a}-a+1+1-\overline{a})[x^0] \\ &= 1. \end{aligned}$$

As a bonus we see that it has a trivial kernel. Next, we'll determine its eigenvalues. We expand the equation $Lf = \lambda f$ to get

$$\begin{aligned} Lf = f + bf_0 &= \lambda f \\ (1-\lambda)f &= -bf_0 \\ f &= \frac{bf_0}{\lambda - 1} \end{aligned}$$

Therefore we determine that f is a constant with respect to x. So the operator's eigenvalues are all real functions $\lambda(a,t) \neq 1$ the eigenfunctions are $f(x,a,t) = \alpha/(\lambda-1)$. This is admitedly not a very interesting situation. What if $\lambda = 1$. Then any function f that is non-zero at $x = 0$ will be an eigenfunction, that is functions of the form $f(x,a,t) = xg(x,a,t)$ where g is finite at $x = 0$.

2.3.2 Operator E

We take $E = 1 - \overline{x}[\overline{x}]$. We have

$$0 = E(f) = f - \overline{x}f_{-1} = 0$$

which means that f is in the kernel of E if $f = c\overline{x}$ for some constant. This shows that E itself is not invertible with kernel generated by $\overline{x}$. Next we determine the

eigenvalues

$$\begin{aligned} Ef &= f - \bar{x}f_{-1} = \lambda f \\ (1-\lambda)f &= f_{-1}\bar{x} \\ f &= \frac{f_{-1}}{x(\lambda-1)} \end{aligned}$$

If $f_{-1} = 0$ then E acts as the identity operator so the eigenvalue is $\lambda = 1$ for every such function. If $f_{-1} \neq 0$ then

$$Ef = \frac{f_{-1}}{x(\lambda-1)} - \frac{1}{x}\frac{f_{-1}}{(\lambda-1)} = 0$$

so the eigenvalue must be 0. Therefore there is a splitting between the functions E acts on. Those without an $\bar{x}$ are preserved with eigenvalue 1. Those who have an $\bar{x}$ and other terms are not preserved. Finally, if f is just such a term, we have eigenvalue 0, of course.

2.3.3 Operator B

The situation here is analogous to the one for L with a different variable and coefficient -1.

2.3.4 Compounded operators

Our first example concerns ES. Since the kernel of E is generated by $\bar{x}$ then a function f is in the kernel of ES if and only if fS is in the kernel of E. That is, the kernel of ES is generated by

$$f = \frac{c}{xS} = \frac{c}{S_{-1} + S_0 x + S_1 x^2}$$

This function can have various forms depending on which coefficients are zero.

Finally we should look at the invertibility and other properties of the operator $1 - tLES$ or even $M - tES$. Okay this is difficult. I'll have to spend more time on this later.

Chapter 3

Higher-dimensional labelled walks

3.1 Using Operators to Derive the Functional Equation

As in the 1D case we'll take the generating function Q, multiply it by tSQ to obtain all the generating function for all walks counted in Q and taken a step further by the step set S. Not all of these are valid so we need to remove the ones that are not. We can apply operator E_1 to retain only those walks that are x_1-valid. We can apply E_2 on the remaining walks to retain only those walks that are x_2-valid as well. We continue like this until E_d is applied and we only have valid walks remaining. Then we can take care of the labelling by applying operator L_1. This takes all the walks counted in tSQ that have ended on the $x_1 = 0$ hyperplane, removes then and re-adds their labelled versions. We can continue applying operators L_i until L_d. In the end we've retained all the valid walks that are also properly re-labelled as they have interacted with the hyperplanes $x_i = 0$. This quatity is $tLE(SQ)$. The only walk from Q that is not present is 1, the empty walk. Therefore we have our functional equation

$$Q = 1 + tLE(SQ)$$

which we can re-write as

$$(1 - tLES)Q = 1$$

We can apply the operator M on this to obtain

$$M(1) = \sum_{I \subseteq [d]} c_I = \overline{a}_{[d]}$$

and

$$(M - tES)Q = \overline{a}_{[d]}$$

This equation will contain many less terms. To see this we take a look at the $L_i E_i$ operator

$$L_i E_i = 1 + b_i[x_i^0] - \overline{x_i}[\overline{x_i}]$$

which increases the number of terms in the equation by a factor of 3. That is we would have 3^d terms relating to SQ by the end of applying LE. Not to mention that each extraction from SQ introduces at most 3 terms. On the other hand once we apply the operator M we introduce 2^d terms relating to Q but then since we only have E applied on SQ we'd only have at most 2^d terms from there. For the case of walks of step size at most 1 and labelling the hyperplane interactions, these terms turn out to be mostly the same. We have a lot fewer terms overall in the latter representation. In this case we can write out the functional equation explicitly. Firstly,

$$MQ = (1 - c_1[x_1^0]) \dots (1 - c_d[x_d^0])Q = \sum_{I \subseteq [d]} (-1)^{|I|} c_I [x_I^{\mathbb{O}}]Q = \sum_{I \subseteq [d]} (-1)^{|I|} c_I Q_I^{\mathbb{O}}$$

and

$$\begin{aligned} ESQ &= (1 - \overline{x}_1[\overline{x}_1]) \dots (1 - \overline{x}_d[\overline{x_d}])(SQ) \\ &= \sum_{I \subseteq [d]} (-1)^{|I|} \overline{x}_I [x_I^{-\mathbb{I}}](SQ) \\ &= \sum_{I \subseteq [d]} (-1)^{|I|} \overline{x}_I S_I^{-\mathbb{I}} Q_I^{\mathbb{O}} \end{aligned}$$

Therefore the functional equation is

$$\sum_{I \subseteq [d]} (-1)^{|I|} (c_I - t\overline{x_I} S_I^{-\mathbb{I}}) Q_I^{\mathbb{O}} = \overline{a_{[d]}}$$

or if we let $K_I = c_I - t\overline{x_I}S_I^{-\mathbb{I}}$ we have

$$\sum_{I\subseteq[d]} (-1)^{|I|} K_I Q_I^{\mathbb{O}} = \overline{a_{[d]}}.$$

3.2 Generalizations to the functional equation

Using the operator notation we can handle generalizations fairly easily. First we can consider a more general step set. Let n_i be non-negative integers for $1 \leq i \leq d$. Then let S be a step set for which no step backtracks more than n_i units in the negative x_i direction. The positive direction is bounded. Then we can write the extraction operators as

$$E_i = 1 - \overline{x}_i[\overline{x}_i] - \ldots - \overline{x}_i^{n_i}[\overline{x}_i^{n_i}]$$

The reason why this works is that this operator must remove all walks that are not x_i-valid and this operator removes all the walks once they step beyond the $x_i = 0$ plane. Of course we could easily generalize to exclude certain step sizes. For example, if the walks have a certain step length, they might be able to jump over wide boundaries (this might be useful in some models). Another example is where we might want to look at all quarter plane walks that don't step at a certain point with coordinates (m,n). Then we can write an operator

$$E(m,n) = 1 - x_1^m x_2^n [x_1^m x_2^n]$$

that we'd apply on SQ to remove those walks that end on (m,n).

Next we take care of labelling. We might want to label the interactions not just with the boundaries but also with specific hyperplanes. For example, if we want to label the boundary interactions and also the interactions with the closest 2 hyperplanes near the $x_i = 0$ hyperplane in the direction of the x_i axis we'd write

$$L_i = 1 + b_{i,0}[x_i^0] + b_{i,1}x_i^1[x_i^1] + b_{i,2}x_i^2[x_i^2]$$

Of course we can also easily label the interactions with specific points. If we want

to label the interactions with point (m,n) with a label $a_{m,n}$ we write

$$L(m,n) = 1 + b_{m,n} x_1^m x_2^n [x_1^m x_2^n]$$

It might be interesting to consider the interactions of quarter plane walks with the line $x_2 = (q/r)x_1$ for some coprime integers q, r. For this we need the label

$$L = 1 + b \sum_{k=0}^{\infty} x_1^{qk} x_2^{rk} [x_1^{qk} x_2^{rk}]$$

This is going to get a little crazy right away so I'm sorry. I just thought of the most general thing I could. All these operators can be thought of as belonging to a family of more generalized operators

$$\Gamma_k(x_1, \ldots, x_d) = \sum_{\substack{I \subseteq [d] \\ |I|=k}} \sum_{J \in \mathbb{Z}^k} \gamma_{I,J} x_I^J [x_I^J]$$

For example the last one can be written as

$$L = 1 + b\Gamma_1(x_1^q x_2^r)$$

for $\gamma_k = 1$ and the first one in the section as

$$E_i = 1 - \Gamma_1(\overline{x}_1)$$

with the sequence $\gamma_k = 1$ for $0 \leq k \leq n_i$. It's more of a thought right now, not something we'll really look into. It suffices to say that all of these operators will commute because there is no mixing of terms in the extractions they perform. We may ask questions about inverses and kernels. The operators E remove terms from the function they are applied on so they are irreversible and so have a non-zero kernel. The labelling operators change terms by a constant so it is enough to remove those constants. For example for the first labelling operator

$$M_i = 1 - c_{i,0}[x_i^0] - c_{i,1} x_i^1 [x_i^1] - c_{i,2} x_i^2 [x_i^2]$$

and then

$$M(m,n) = 1 - c_{m,n} x_1^m x_2^n [x_1^m x_2^n]$$

This is because $-c = 1/a-$ so the operator removes the old term and replaces it with one multiplied by $1/a$. This discussion allows us to still write the functional equation as

$$(1 - tLES)Q = 1$$

and upon applying M we have

$$(M - tES)Q = M(1)$$

where $M(1)$ will depend only on the labelling of contacts with the hyperplanes $x_i = 0$ and not any other points. We could write out explicit equations for some operators.

3.3 Coefficient Extraction

We can perform coefficient extraction like in the 1-dimensional case. The operator $[x_i^k]$ will commute with all M_j, E_j with $j \neq i$ so that we only need to consider the action on the operators M_i and E_i. We've already done so in the 1-dimensional case. The zeroth power extraction yields

$$[x_i^0]M_i = [x_i^0] - [x_i^0]c_i[x_i^0] = [x_i^0] - c_i[x_i^0] = \overline{a_i}[x_i^0]$$
$$[x_i^0]E_i = [x_i^0] - [x_i^0]\overline{x_i}[\overline{x_i}] = [x_i^0]$$

and the k-th power extraction for $k \geq 1$ yields

$$[x_i^k]M_i = [x_i^k] - [x_i^k]c_i[x_i^0] = [x_i^k]$$
$$[x_i^k]E_i = [x_i^k] - [x_i^k]\overline{x_i}[\overline{x_i}] = [x_i^k]$$

So, for $k \geq 0$ the operator $[x_i^k]$ eliminates both M_i and E_i from the picture while for $k = 0$ it removes E_i but leaves a residual $\overline{a_i}$ behind for M_i. Therefore if we extract

$[x_I^{\mathbb{O}}]$ from the functional equation we have

$$\begin{aligned}[x_I^{\mathbb{O}}]a_{[d]} = \overline{a}_{[d]} &= [x_I^{\mathbb{O}}](M_{[d]} - tE_{[d]}S)Q \\ &= \left(\overline{a}_I M_{[d]\setminus I}[x_I^{\mathbb{O}}] - tE_{[d]\setminus I}[x_I^{\mathbb{O}}]S\right)Q \\ &= [x_I^{\mathbb{O}}](\overline{a}_I M_{[d]\setminus I} - tE_{[d]\setminus I}S)Q\end{aligned}$$

so that

$$\overline{a}_{[d]\setminus I} = [x_I^{\mathbb{O}}](M_{[d]\setminus I} - a_I tE_{[d]\setminus I}S)Q$$

Written in another way this becomes

$$\overline{a}_I = [x_{[d]\setminus I}^{\mathbb{O}}](M_I - a_{[d]\setminus I}tE_I S)Q$$

and we can use whichever notation is more useful at the time. The latter notation underscores the idea that when we extract coefficients for walks that end at some hyperplane intersection we are having a reduced version of the functional equation at the cost of an added a term. When we extract a general term we can use the operator $[x_I^{\mathbb{O}} x_J^P]$ where P is a set of powers greater than 0 and $I \cap J = \emptyset$, $|P| \geq 1$ and $I \cup J = K$. We find

$$\begin{aligned}[x_I^{\mathbb{O}} x_J^P]a_{[d]} = 0 &= [x_I^{\mathbb{O}} x_J^P](M_{[d]} - tE_{[d]}S)Q \\ &= [x_I^{\mathbb{O}} x_J^P](\overline{a}_I M_{[d]\setminus K} - tE_{[d]\setminus K}S)Q\end{aligned}$$

which we can write as

$$[x_I^{\mathbb{O}} x_J^P](M_{[d]\setminus K} - a_I tE_{[d]\setminus K}S)Q = 0$$

If I is empty then

$$[x_J^P](M_{[d]\setminus J} - tE_{[d]\setminus J}S)Q = 0$$

The problem with a lot of things in this section is that they are not really practical beyond providing a shortcut for some of the calculations so we don't have to do them by hand every time. Also they provide a broader idea of having multiple kernels in one problem. Perhaps things will be clearer when we specialize to the

2-dimensional problem.

3.4 Further functional equations

We can try going a little further in the coefficient extraction to come up with an equation that looks like the original functional equation. We know that

$$[x_I^{\mathbb{O}}]SQ = S_I^{\mathbb{O}} Q_I^{\mathbb{O}} + R_I$$

where R_I is a collection of $S_I^{-P} Q_I^P$ terms that involves sets P whose terms are not all zero. We can use this inside the functional equation

$$\begin{aligned}
\overline{a}_{[d]\setminus I} &= M_{[d]\setminus I}[x_I^{\mathbb{O}}]Q - a_I t E_{[d]\setminus I}[x_I^{\mathbb{O}}](SQ) \\
&= M_{[d]\setminus I} Q_I^{\mathbb{O}} - a_I t E_{[d]\setminus I}(S_I^{\mathbb{O}} Q_I^{\mathbb{O}} + R_I) \\
&= (M_{[d]\setminus I} - a_I t E_{[d]\setminus I} S_I^{\mathbb{O}}) Q_I^{\mathbb{O}} - a_I t E_{[d]\setminus I} R_I
\end{aligned}$$

so we can re-write the functional equation as

$$(M_{[d]\setminus I} - a_I t E_{[d]\setminus I} S_I^{\mathbb{O}}) Q_I^{\mathbb{O}} = a_I(\overline{a}_{[d]} + t R_I')$$

where $R_I' = E_{[d]\setminus I} R_I$. We can calculate the following terms

$$M_{[d]\setminus I} Q_I^{\mathbb{O}} = \sum_{J \subseteq [d]\setminus I} (-1)^{|J|} c_J Q_{I,J}^{\mathbb{O},\mathbb{O}}$$

and

$$a_I t E_{[d]\setminus I} S_I^{\mathbb{O}} Q_I^{\mathbb{O}} = \sum_{J \subseteq [d]\setminus I} (-1)^{|J|} a_I t \overline{x}_J S_{I,J}^{\mathbb{O},-\mathbb{I}} Q_{I,J}^{\mathbb{O},\mathbb{O}}$$

so the functional equation can be written as

$$\sum_{J \subseteq [d]\setminus I} (-1)^{|J|} (c_J - a_I t \overline{x}_J S_{I,J}^{\mathbb{O},-\mathbb{I}}) Q_{I,J}^{\mathbb{O},\mathbb{O}} = a_I(\overline{a}_{[d]} + t R_I')$$

This suggests the notation

$$K_J(I) = c_J - a_I t \overline{x}_J S_{I,J}^{\mathbb{O},-\mathbb{I}}$$

where when $I = \emptyset$, $K_J(I) = K_J$ from before. This allows us to write the 2^d functional equations in the simplified form

$$\sum_{J \subseteq [d] \setminus I} (-1)^{|J|} K_J(I) Q_{I,J}^{\mathbb{O},\mathbb{O}} = \overline{a}_{[d]\setminus I} + a_I t R'_I$$

where the R'_I term involves only higher order coefficient extractions from Q, that is not all the powers are zero. Now, to calculate the R'_I terms we first determine what R_I is. It is a sum of terms $S_I^{-P} Q_I^P$ for various nonnegative P with size $|I|$ whose elements are not all 0. To calculate R'_I we apply the operator $E_{[d]\setminus I}$ on R_I. We can try calculating some of these quantities. A key observation is that the terms Q_I^P are power series in the remaining variables so they don't have non-negative coefficients. Let $sq = S_I^{-P} Q_I^P$. Then

$$\begin{aligned} E_j sq &= sq - \overline{x}_j (sq)_j^{-1} \\ &= sq - \overline{x}_j (s_j^{-1} q_j^0 + s_j^0 q_j^{-1} + s_j^1 q_j^{-2}) \\ &= sq - \overline{x}_j s_j^{-1} q_j^0 \end{aligned}$$

Hence

$$E_K sq = \sum_{J \subseteq K} (-1)^{|J|} \overline{x}_J s_J^{-\mathbb{I}} q_J^{\mathbb{O}}$$

Finally we can write

$$\begin{aligned} R'_I &= E_{[d]\setminus I} R_I \\ &= \sum_{P > \mathbb{O}} E_{[d]\setminus I} S_I^{-P} Q_I^P \\ &= \sum_{P > \mathbb{O}} \sum_{J \subseteq [d]\setminus I} (-1)^{|J|} S_{I,J}^{-P,-\mathbb{I}} Q_{I,J}^{P,\mathbb{O}} \\ &= \sum_{J \subseteq [d]\setminus I} (-1)^{|J|} \sum_{P > \mathbb{O}} S_{I,J}^{-P,-\mathbb{I}} Q_{I,J}^{P,\mathbb{O}} \end{aligned}$$

This is not a very nice formula to use but it should do for now.

3.5 Using the formal inverse for iteration

In a previous subsection we obtained the equation

$$(1 - tLES)Q = 1$$

Let $D = LE$. Then we can think of the formal inverse of this operator against some norm $||tDS||$ or even formally. Formally

$$(1 - tDS)^{-1} = \frac{1}{1 - tDS} = \sum_{k=0}^{\infty} t^k (DS)^k$$

so we can write the solution for Q as

$$Q = \sum_{k=0}^{\infty} t^k (DS)^k (1)$$

where $(DS)^k(1) = DS(\ldots(DS(1))\ldots)$. This allows us to compute the series expansion for Q directly using an iteration. The next coefficient is the previous one with DS applied to it. It can be done fairly efficiently on a computer. Interestingly enough this is just a formalization of the brute force method of determining the functional equation - by taking a previous set of walks, extending them by 1 step, removing the ones that fall outside of the quarter plane and finally labelling the ones that have interacted with the hyperplanes $x_i = 0$ in the process. We could attempt to write a closed form for the operator application but it becomes messy fairly fast, even for one dimension.

Chapter 4

Specializing to 2 dimensions

4.1 Introduction

In a lot of ways the 2-dimensional problem looks like the 1-dimensional problem but there are major differences in the inherent complexity of the work that needs to be done. We have the functional equations

$$\begin{aligned} KQ - K_1Q_1^0 - K_2Q_2^0 + K_{1,2}Q_{1,2}^{0,0} &= \overline{a}_1\overline{a}_2 \\ K(1)Q_1^0 - K_2(1)Q_{1,2}^{0,0} &= \overline{a}_2 + a_1tR_1' \\ K(2)Q_2^0 - K_1(2)Q_{1,2}^{0,0} &= \overline{a}_1 + a_2tR_2' \\ K(1,2)Q_{1,2}^{0,0} &= 1 + a_1a_2tR_{1,2}' \end{aligned}$$

Here

$$\begin{aligned} R_1 &= S_1^{-1}Q_1^1 \\ R_2 &= S_2^{-2}Q_2^1 \\ R_{1,2} &= S_{1,2}^{0,-1}Q_{1,2}^{0,1} + S_{1,2}^{-1,0}Q_{1,2}^{1,0} + S_{1,2}^{-1,-1}Q_{1,2}^{1,1} \end{aligned}$$

with

$$R_1' = S_1^{-1}Q_1^1 - \overline{x}_2 S_{1,2}^{-1,-1}Q_{1,2}^{1,0}$$
$$R_2' = S_2^{-1}Q_2^2 - \overline{x}_1 S_{1,2}^{-1,-1}Q_{1,2}^{0,1}$$
$$R_{1,2}' = S_{1,2}^{0,-1}Q_{1,2}^{0,1} + S_{1,2}^{-1,0}Q_{1,2}^{1,0} + S_{1,2}^{-1,-1}Q_{1,2}^{1,1}$$

We can write the functional equations in the explicit form

$$(1-tS)Q - (c_1 - t\overline{x_1}S_1^{-1})Q_1^0 - (c_2 - t\overline{x_2}S_2^{-1})Q_2^0 + (c_1c_2 - t\overline{x_1x_2}S_{1,2}^{-1,-1})Q_{1,2}^{0,0} = \overline{a_1a_2} \tag{4.1}$$

$$(1-ta_1S_1^0)Q_1^0 - (c_2 - ta_1\overline{x_2}S_{1,2}^{0,-1})Q_{1,2}^{0,0} - a_1t(S_1^{-1}Q_1^1 - \overline{x_2}S_{1,2}^{-1,-1}Q_{1,2}^{1,0}) = \overline{a_2} \tag{4.2}$$

$$(1-ta_2S_2^0)Q_2^0 - (c_1 - ta_2\overline{x_1}S_{1,2}^{-1,0})Q_{1,2}^{0,0} - a_2t(S_2^{-1}Q_2^1 - \overline{x_1}S_{1,2}^{-1,-1}Q_{1,2}^{0,1}) = \overline{a_1} \tag{4.3}$$

$$(1-a_1a_2tS_{1,2}^{0,0})Q_{1,2}^{0,0} - ta_1a_2t(S_{1,2}^{-1,-1}Q_{1,2}^{1,1} + S_{1,2}^{-1,0}Q_{1,2}^{1,0} + S_{1,2}^{0,-1}Q_{1,2}^{0,1}) = 1 \tag{4.4}$$

where the R_I' terms are written on the left to conserve space. We can already see that we are working in a much richer space than in the 1-dimensional case. We'll mostly work on walks with step sets that don't allow staying in one spot ($S_{1,2}^{0,0} = 0$). Sometimes we might though.

4.2 Investigating relationships between G, G_1, G_2 and $G_{1,2}$

In this work we'll focus on investigating the properties of the function $G(a_1,a_2) = Q_{1,2}^{0,0}(a_1,a_2)$. Let $G_1 = G(1,a_2)$, $G_2 = G(a_1,1)$ and $G_{1,2} = G(1,1)$. We could also see these as coefficient extractions of G with respect to b_1, b_2 and b_1b_2, i.e. $G_{1,2} = [b_1^0b_2^0]G$. In previous works it has been shown that there is a relationship between these quantities depending on the particular step set S. In some cases (like diagonal walks $S = (x_1+\overline{x_1})(x_2+\overline{x_2})$) we have a linear relationship between these quantities. In other cases (like cartesian walks $S = x_1+\overline{x}_1+x_2+\overline{x}_2$) we don't have such a linear relationship but we have a rational one, that is G is a rational function of G_1, G_2 and $G_{1,2}$. The aim of the followin sections is to investigate when such

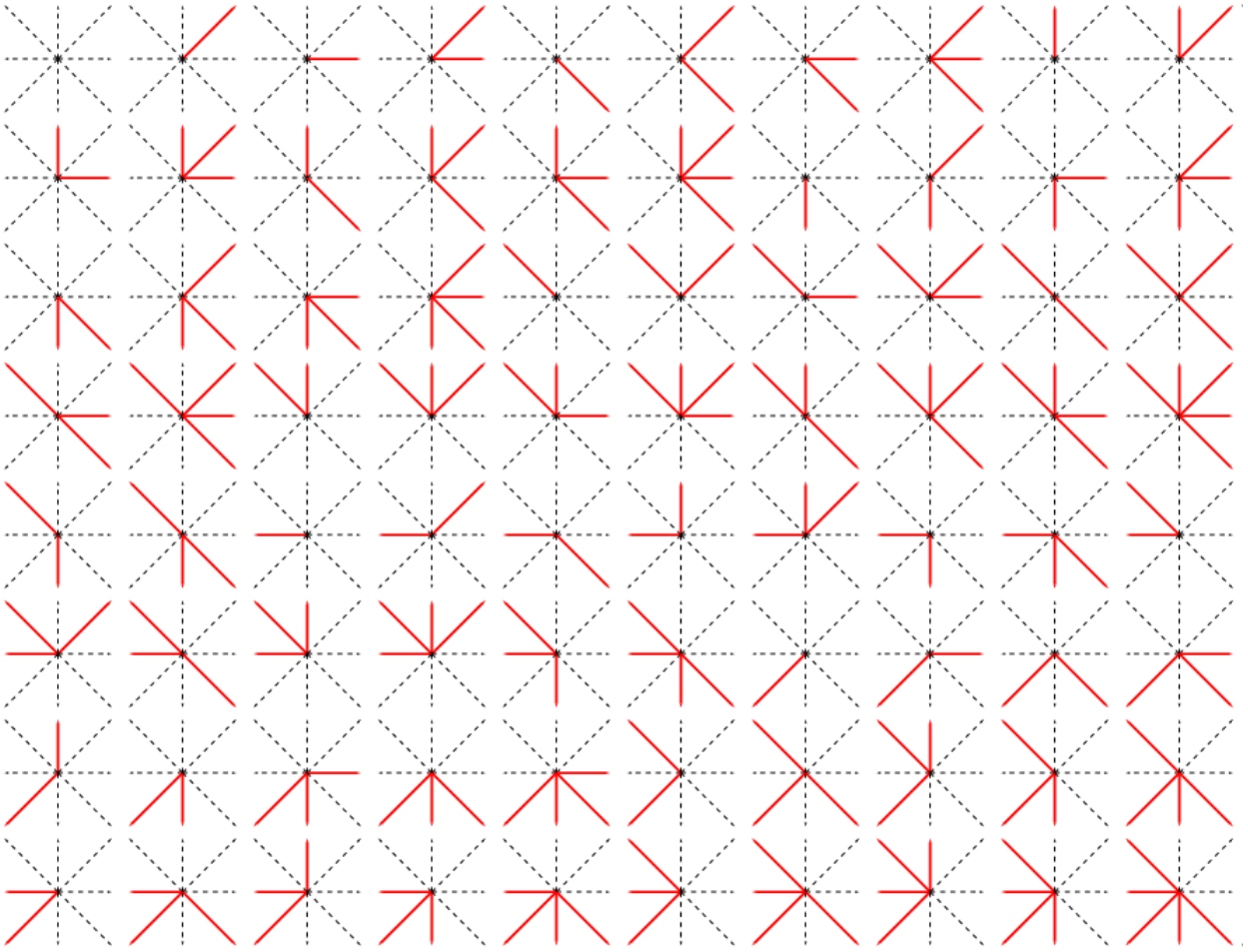
relationships exist, when se have a relationship of that type and whether we have other cases.

The work can be split into several cases - walks that never leave the origin and ones that leave but never return to the origin. In the first case the relationship is trivially $G - G_1 - G_2 + G_{1,2} = 0$ because $Q = 1$ itself and so $G = 1$. In the second case Q may be more complicated but again $G = 1$ trivially since no walks end at the origin. This case satisfies the same relationship. However, once we move beyond this stage we need to consider the problem more carefully.

In any of the simple cases we can write $BG = 0$ and sometime $BQ = 0$ where $B = B_1 B_2$ from before. We'll investigate the when we'll have a relationship like $B(FQ) = 0$, $B(FG) = 0$ or $B(FG)$ is constant for some F's.

4.3 Simple cases: Walks never leaving the origin and walks leaving the origin but never returning

Here are all the easy cases of step sets that lead to walks that either don't leave the origin or leave it and never come back. We're not allowing resting steps here. There are 80 such sets among the 256 non-resting possible cases. Of course we could use symmetries to eliminate cases but for now this was an easier thing to do.



There are $2^5 = 32$ types of walks that don't leave the origin. Those are walks with steps among the set {NW,W,SW,S,SE}. Next we look at the walks that leave the origin but never come back. There is the case of $S = \{NE\}$ that never comes back to any axis. Let's look at those walks that never come back to the x_1 axis. We have some steps that go in the positive x_2 direction, that is one of {N, NE, E} together with some steps from {NW, W, SW}. If we were to include any other steps we would come back to the x_1 axis. If we only have a N step, we have 8 new cases, if we have only a NE step we have 3 new cases and if we have both, we have 4 new cases. This gives a total of 15 new cases. There's an equivalent number of walks that leave the x_2 axis and never come back to it. The running total is $32+1+15+15 = 63$.

To finish off we have 17 cases of walks that leave the origin, eventually interact with both axes but never come back to the origin. For this we must only include steps in such a way that we cannot pick a selection of steps that will decrease both the x_1 and x_2 coordinates. We have to pick a subset of steps from {N, NE, E} so the only other steps allowed are NW and SE. The new cases coming from {N, NE, E}, {N,E}, {N, NE }, {NE, E}, {N}, {NE}, {E} are 4, 4, 2, 2, 1, 2, 2 respectively.

Before we consider walks that come back to the origin we'll look at the behavior Q depending on whether S factors in some nice way or not.

4.4 Simple case: Factoring S into a product

There's something we can say about the case where S factors as $S = S_1(x_1)S_2(x_2)$. We have

$$(DS)^k = (D_1D_2S_1S_2)^k = (D_1S_1)^k(D_2S_2)^k$$

since the D_1 and D_2 operators commute and the D_iS_i operators also commute since they deal with different variables. The functional equation for Q is

$$(1 - tDS)Q = 1$$

We can compute Q directly via the iteration as

$$Q = \sum_{k=0}^{\infty} t^k(DS)^k = \sum_{k=0}^{\infty} t^k(D_1S_1)^k(D_2S_2)^k = Q_1(x_1, a_1, t) \odot_t Q_2(x_2, a_2, t)$$

where $\odot_t$ denotes the Hadamard product of two functions with respect to variable t. From now on we'll drop the subscript t unless it is needed. Here Q_i is given by

$$Q_i = \sum_{k=0}^{\infty} t^k(D_iS_i)^k$$

and by reversing the iteration we obtain the functional equation

$$(1 - tD_iS_i)Q_i = 1$$

In other words, $Q_i(x_i, a_i, t)$ is the generating function for a 1-dimensional problem with Laurent step polynomial $S_i(x_i)$. In the case of a factoring of this sort, we are able to glue the 2-dimensional solution out of two 1-dimensional solutions. If we want to look at the generating function for quarter plane walks of length k it is enough to look at the product of two generating functions for one-dimensional walks of length k. Also, we can write

$$Q = Q_1 \odot Q_2, \quad \widehat{Q} = \widehat{Q}_1 \odot Q_2$$
$$\widehat{Q} = Q_1 \odot \widehat{Q}_2, \quad \widehat{\widehat{Q}} = \widehat{Q}_1 \odot \widehat{Q}_2$$

Therefore

$$B_1 B_2 Q = Q - \widehat{Q} - \widehat{Q} + \widehat{\widehat{Q}} = (Q_1 - \widehat{Q}_1) \odot (Q_2 - \widehat{Q}_2)$$
$$B_1 B_2 (Q_1 \odot Q_2) = (B_1 Q_1) \odot (B_2 Q_2)$$

In general if we are applying operators that don't involve t to a Hadamard product with respect to t we can treat the operator as a constant with respect to t and apply it on whichever term we would like to. If it applies to both terms then we have to apply it to their mixture in a non-trivial way. For example, each of the Q_i's solves a 1-dimensional problem and therefore we have a relationship of the type $B_i(f_i Q_i) = g_i$ for some f_i and g_i. Then we can write

$$B_1 B_2 (f_1 Q_1 \odot f_2 Q_2) = (B_1 f_1 Q_1) \odot (B_2 f_2 Q_2) = g_1 \odot g_2$$

but so far this is not a relationship of the type $B_1 B_2 (fQ) = g$ since taking out the functions f_1 and f_2 from within the Hadamard product is non-trivial in general. One might be tempted to say that $f = f_1 \odot f_2$ but that would be incorrect. It would be neat to see if there is a way to find f in terms of f_1 and f_2.

There is a way to see the Hadamard product solution without the iteration method. Suppose that Q_i is a solution to the functional equation $(M_i - tE_i S_i)Q_i = \overline{a}_i$ for $i = 1$ and $i = 2$. Then we claim that $Q = Q_1 \odot Q_2$ is a solution to the functional

equation $(M - tES)Q = \overline{a}_1\overline{a}_2$. To this end we calculate

$$\begin{aligned}(M - tES)Q &= M(Q_1 \odot Q_2) - tES(Q_1 \odot Q_2) \\ &= (M_1Q_1 \odot M_2Q_2) - t((E_1S_1Q_1) \odot (E_2S_2Q_2))\end{aligned}$$

Then we substitute $M_iQ_i = \overline{a}_i + tE_iS_iQ_i$ into the first term to obtain

$$\begin{aligned}(M_1Q_1 \odot M_2Q_2) &= (\overline{a}_1 + tE_1S_1Q_1) \odot (\overline{a}_2 + tE_2S_2Q_2) \\ &= \overline{a}_1 \odot \overline{a}_2 + \overline{a}_1 \odot (tE_2S_2Q_2) + \overline{a}_2 \odot (tE_1S_1Q_1) + (tE_1S_1Q_1) \odot (tE_2S_2Q_2))\end{aligned}$$

We have

$$\overline{a}_i \odot (tE_jS_jQ_j) = \overline{a}_i[t^0]tE_jS_jQ_j = 0$$

since we are dealing with power series in t. Therefore

$$(M_1Q_1 \odot M_2Q_2) = \overline{a}_1\overline{a}_2 + t((E_1S_1Q_1) \odot (E_2S_2Q_2))$$

and upon substitution we are left with $\overline{a}_1\overline{a}_2$ which is what we needed in the first place.

One may also ask whether we can do something in the case where S factors and the terms share variables.

This factoring also generalizes to higher dimensions. If S factors into a product of k factors $S_1, \ldots, S_k$ where no two of which share any variables then the problem of finding Q boils down to solving k lower-dimensional problems for functions $Q_1, \ldots, Q_k$, naming variables appropriately and taking the Hadamard product of the functions. That is

$$Q = Q_1 \odot \cdots \odot Q_k$$

The more lower-dimensional problems that we solve, the more higher dimensional problems we can solve. Although it appears that the interesting higher dimensional problems that can be solved in this manner will have a lower and lower proportion among all higher dimensional problems.

4.5 Kernel Group Considerations

It is important to find more functional equations if we can. One way to do this is by applying a transformation on the pair (x_1, x_2) that leaves as many parts of the functional equation as we can but still adding some information in the form of a new equation. One way to do this is to cosider mapping x_1 and x_2 to some quantities separately and then consider the full group generated by such maps. Since we usually eliminate the kernel in front of Q it is useful to consider transformations that preserve the kernel. We'll call this group $G(K)$, although it won't be the full group of rational transformations preserving the kernel. For example $(x_1, x_2) \mapsto (x_2, x_1)$ will not be in the group. Any transformation on (x_1, x_2) that preserves the kernel $K = 1 - tS$ must necessarily preserve the step polynomial S. Here

$$S = \alpha\overline{x} + \beta + \gamma x$$

where $x = x_1$ or x_2 and α, β, γ change correspondingly. We consider sending $x \mapsto X$ and solve for X to determine the possible transformations we can perform. That is we have

$$\begin{aligned} \alpha\overline{x} + \gamma x &= \alpha\overline{X} + \gamma X \\ \alpha(\overline{x} - \overline{X}) &= \gamma(X - x) \\ \alpha\frac{X - x}{Xx} &= \gamma(X - x) \end{aligned}$$

We have cases

- $\alpha = 0, \gamma = 0$. The equation is then trivially true and any transformation preserves the kernel although this is not very useful though.
- $\alpha = 0, \gamma \neq 0$. Then $X = x$ and only the identity transformation preserves the kernel.
- $\alpha \neq 0, \gamma = 0$. Here $X = x$ as well.
- $\alpha \neq 0, \gamma \neq 0$. Here we either have $X = x$ or $X \neq x$ and $X = \overline{x}\alpha/\gamma = \overline{x}S_{-1}/S_1$

In conclusion, either every transformation works and we don't have an interesting case or we have the identity transformation plus a possible second one. The

last transformation creates a non-trivial group and we can have groups of various orders. Let

$$g_1 : x_1 \mapsto \overline{x_1}\frac{S_1^{-1}}{S_1^1}, \quad g_2 : x_2 \mapsto \overline{x_2}\frac{S_2^{-1}}{S_2^1}$$

whenever the S quantities are non-zero. Note that $g_i(g_i(x_1)) = x_1$, that is they are involutions. We have three cases for each x

- 1: Any transformation can be applied
- 2: We only have the identity transformation
- 3: We have two transformations

We'll combine cases for analysis in the form "case $i.j$" to mean we have case i for x_1 and case j for x_2.

Case 1.1: Here we can apply any transformation on both x_1 and x_2. From the transformation analysis this means that $S_1^{-1} = S_1^1 = S_2^{-1} = S_2^1 = 0$ and thus means that we have no steps that go away from the origin. Therefore $S = S_{1,2}^{0,0}$. Thus we have only two models - no steps or a resting step. (I should include little diagrams with arrows like before).

Case 1.2 : Here any transformation applies on x_1, so $S_1^{-1} = S_1^1 = 0$ so the model has no steps that step away from the origin in the x_1-direction. Since the only transformation on x_2 is the identity transformation we either have $S_2^{-1} = 0$ or $S_2^1 = 0$ in which case the model consists of only an up step or a down step, together with a possible resting step. Similarly, the 2.1 case involves a single x_1 or $\overline{x_1}$ step with a possible resting step.

Case 1.3: (and 3.1): This case is similar to the previous one except that now we have two transformations for x_2 and this means $S_2^{-1}, S_2^1 \neq 0$. This means we must have either both vertical steps x_2 and $\bar{x}_2$ in the 1.3 case or two horizontal steps x_1 and $\bar{x}_1$ in the 3.1 case.

Case 2.2: In this case both transformations are the identity transformations. Therefore exactly one of S_1^{-1} and S_1^1 is zero and exactly one of S_2^{-1} and S_2^1 is zero. WLOG, assume that $S_1^1 \neq 0, S_2^1 \neq 0$ (the other models can be obtained by

a rotation around the origing of 90, 180 and 270 degrees). The possible models are (I should include the drawings): {N,E,NE}, {N,NE}, {E,NE}, {N,E}, {NE}. In other words, the subsets of {N,E,NE} except the ones going either only in the N direction or only in the E direction. This is a case of group $G(K)$ or order 1.

Case 2.3: In this case there is only one transformation for x_1 and two for x_2. This means $S_1^{-1} = 0$ (positive case) or $S_1^1 = 0$ (negative case). Also $S_2^{-1}, S_2^1 \neq 0$ so there must be at least one step going in the positive x_2 direction and one in the negative x_2 direction. We have 17 cases (and another 17 including a resting step). Of course in the 3.2 case we have the meaning of x_1 and x_2 reversed.This is a case of group $G(K)$ of order 2.

Case 3.3: This is the most interesting case with two transformations for x_1 and x_2 each. This means that there must be at least one step going in a positive and one going in a negative coordinate for each of x_1 and x_2. There are a lot of such cases (in the hundreds). Previous results show that the group $G(K)$ has orders 4, 6, 8 and infinity depending on the step set in question.

Chapter 5

Solving basic 2D cases

5.1 Cases 1.1, 1.2, 1.3

A quick glance at the step models involved shows that we are dealing with essentially one-dimensional problems. This means they can be solved easily explicitly. One can also note that these models are all factorable

$$S(x_1,x_2) = 1 \cdot S(x_2)$$

Both Q_1 and Q_2 are functions we solved for in the one-dimensional case

$$Q_1(x_1,a_1,t) = \frac{1}{1-a_1 t}, \quad Q_2(x_2,a_2,t) = q(x_2,a_2,t)$$

for some solution q from that section. Therefore

$$Q(x_1,x_2,a_1,a_2,t) = Q_1(x_1,a_1,t) \odot Q_2(x_2,a_2,t) = \frac{1}{1-a_1 t} \odot q(x_2,a_2,t) = q(x_2,a_2,a_1 t)$$

Similarly in the 2.1, 3.1 cases

$$Q(x_1,x_2,a_1,a_2,t) = Q_1(x_1,a_1,t) \odot Q_2(x_2,a_2,t) = q(x_1,a_1,t) \odot \frac{1}{1-a_2 t} = q(x_1,a_1,a_2 t)$$

and this completes the solution. What about the relationship with the $B_1 B_2$ transformation? If $f(a,t)$ is the function that goes together with $q(x,a,t)$ then the function

$F(a_1,a_2,t)$ that goes together with $Q(x_1,x_2,a_1,a_2,t)$ is given by

$$F(a_1,a_2,t) = f(a_2,a_1t)$$

Then we have $B_2(FQ) = 0$ and $B_1B_2(FQ) = 0$.

5.2 Case 2.2 - group of order 1

A quick glance at the step models reveals that this case is mostly trivial. We have a lot of models that never leave the origin and those have a trivial solution

$$Q(x_1,x_2,a_1,a_2) = \frac{1}{1-a_1a_2S_{1,2}^{0,0}t}.$$

We also have a lot of models that are essentially one-dimensional with a single step going along an axis and some steps going off-axis where the latter steps can never be taken by a random walk in this model. They are essentially cases already covered in the previous section. Suppose that the single step is taken in the positive x_1 direction. This is equivalent to the $s_{-1} = 0$ case in 1D with function

$$q(x,a,t) = \frac{1-s_0t}{(1-st)(1-as_0t)} = \frac{1-s_0t}{(1-x_1s_1t)(1-as_0t)}$$

Then we have

$$Q(x_1,x_2,a_1,a_2,t) = Q_1(x_1,a_1,t) \odot Q_2(x_2,a_2,t) = q(x_1,a_1,t) \odot \frac{1}{1-a_2t} = q(x_1,a_1,a_2t)$$

and therefore

$$Q(x_1,x_2,a_1,a_2,t) = \frac{1-a_2S_{1,2}^{0,0}t}{(1-(x_1+S_{1,2}^{0,0})a_2t)(1-a_1a_2S_{1,2}^{0,0}t)}$$

The interesting models in case 2.2 are those with non-negative steps that we listed. Here

$$S = \alpha x_1 + \beta x_2 + \gamma x_1x_2 + \delta$$

The functional equation and its zero extractions are

$$\begin{aligned}(1-tS)Q - c_1Q_1^0 - c_2Q_2^0 + c_1c_2Q_{1,2}^{0,0} &= \overline{a_1a_2}\\ (1-ta_1(\delta+\beta x_2))Q_1^0 - c_2Q_{1,2}^{0,0} &= \overline{a_2}\\ (1-ta_2(\delta+\alpha x_1))Q_2^0 - c_1Q_{1,2}^{0,0} &= \overline{a_1}\\ (1-a_1a_2t\delta)Q_{1,2}^{0,0} &= 1\end{aligned}$$

We solve the problem by a simple back-substitution of the variables.

$$\begin{aligned}Q_{1,2}^{0,0} &= \frac{1}{1-a_1a_2\delta t}\\ Q_1^0 &= \frac{\overline{a}_2}{1-ta_2S_1^0}\left(1+\frac{b_2}{1-a_1a_2\delta t}\right) = \frac{1-a_1\delta t}{(1-ta_1S_1^0)(1-a_1a_2\delta t)}\\ Q_2^0 &= \frac{\overline{a}_1}{1-ta_1S_2^0}\left(1+\frac{b_1}{1-a_1a_2\delta t}\right) = \frac{1-a_2\delta t}{(1-ta_2S_2^0)(1-a_1a_2\delta t)}\\ Q &= \frac{1}{1-tS}(\overline{a}_1\overline{a}_2 + c_1Q_1^0 + c_2Q_2^0 - c_1c_2Q_{1,2}^{0,0})\\ &= \frac{1}{1-tS}\left(\overline{a}_1\overline{a}_2 + \frac{1}{1-ta_1a_2\delta}\left(c_1\frac{1-a_1\delta t}{1-ta_1S_1^0} + c_2\frac{1-a_2\delta t}{1-ta_2S_1^0} - c_1c_2\right)\right)\\ &= \frac{1}{1-tS}\left(\overline{a}_1\overline{a}_2 + \frac{1}{1-a_1a_2\delta t}(c_1f_1(a_1,t) + c_2f_2(a_2,t) - c_1c_2)\right)\end{aligned}$$

where the last line explicitly shows that Q is composed of functions that depend only on one of a_1, a_2. If we want to find a relation of the type $B(FQ) = 0$ then we must first find a suitable function F. First of all the $1-tS$ part doesn't play a role since it doesn't depend on a_1 or a_2. Then later the variables c_1 and c_2 will vanish when extracting the coefficients of b_1^0 and b_2^0. We'll now show that the function

$$F(a_1,a_2,t) = 1-a_1a_2\delta t$$

works exactly. First

$$B((1-a_1a_2\delta t)\overline{a}_1\overline{a_2}) = B(\overline{a}_1\overline{a}_2 - t\delta) = B(\overline{a}_1\overline{a}_2) - B(t\delta) = \overline{a}_1\overline{a}_2 - \overline{a}_2 - \overline{a}_1 + 1 = c_1c_2$$

Then

$$B(c_1f_1 + c_2f_2 - c_1c_2) = c_1f_1 + c_2f_2 - c_1c_2 - c_2f_2 - c_1f_1 = -c_2c_2$$

Putting the two quantities together along with dividing by $1-tS$ shows that $B(FQ) = 0$. Of course this works for $G = Q^{0,0}_{1,2}$ too.

5.3 Case 2.3 - group of order 2

There are two possibilities here - positive and negative. In the negative case $S^1_1 = 0$ and the steps have a non-positive x_1 component. Due to the fact that a lot of steps are not going to be taken by random walks in those models, all of these models reduce to the cases of S consising of a rest step and maybe a single step in a positive direction. This takes care of 34 models (or 68 if we include a rest step). The interesting models are those with a positive x_1 component, or in other words, models where $S_1^{-1} = 0$. The calculations for these models become increasingly more complicated. The functional equation and its extracted forms read as

$$\begin{aligned}
(1-tS)Q - c_1Q^0_1 - (c_2 - t\bar{x}_2S_2^{-1})Q^0_2 + c_1c_2Q^{0,0}_{1,2} &= \overline{a_1a_2} \\
(1-ta_1S^0_1)Q^0_1 - (c_2 - ta_1\bar{x}_2S^{0,-1}_{1,2})Q^{0,0}_{1,2} &= \overline{a_2} \\
(1-ta_2S^0_2)Q^0_2 - c_1Q^{0,0}_{1,2} - a_2tS_2^{-1}Q^1_2 &= \overline{a_1} \\
(1-a_1a_2tS^{0,0}_{1,2})Q^{0,0}_{1,2} - ta_1a_2S^{0,-1}_{1,2}Q^{0,1}_{1,2} &= 1
\end{aligned}$$

This is starting to look somewhat daunting. However, we can identify a subproblem that is essentially one-dimensional and therefore already solved. For this we look at the second and fourth equations with the substitution $a_i = a, c_i = c, x_2 = x, ta_1 = \tau, S^0_1 = T, Q^0_1(x_2) = q(x)$. These equations become

$$\begin{aligned}
(1-\tau T)q - (c - \tau\bar{x}T_{-1})q_0 &= \overline{a} \\
(1-a\tau T_0)q_0 - a\tau S^{0,-1}_{1,2}q_1 &= 1
\end{aligned}$$

where $q_0 = q(a,\tau)$. This is the one-dimensional problem again. The solution here is $q(x,a,\tau)$ so the original problem has a solution

$$Q_1^0(x_2) = q(x_2, a_2, a_1 t), \quad Q_{1,2}^{0,0} = q_0(a_2, a_1 t).$$

The reason the method worked in this case is that we have steps that keep a random walk staying on the x_2 axis and steps that take it away from the x_2 but no steps that have it return to it. This is similar to the case where a random walk leaves the x_2 axis never to come back. Essentially the problem splits into a one-dimensional one regarding the x_2 axis and a further 2D one once we step off the axis. We re-write the functional equation as

$$(1-tS)Q - (c_2 - t\bar{x}_2 S_2^{-1})Q_2^0 = \overline{a}_1 \phi(x_2) = \overline{a_1}(\overline{a}_2 + b_1 Q_1^0 - b_1 c_2 Q_{1,2}^{0,0})$$

Here we can treat x_2 as a constant and solve for a one-dimensional problem with x_1 as a variable. We can perform the same analysis as in the the one-dimensional case except where we had $\overline{a}_1$ we would now have $\overline{a}_1\phi(x_2)$. Therefore we can solve for Q explicitly. However I haven't done this yet so I'll leave this for later. Curiously enough this analysis does not use the fact that we are dealing with a group of order 2 anymore than the one-dimensional case uses the fact that $|G(K)| = 2$ for $K = 1 - t(\bar{x} + x)$. Interestingly we can try re-doing the problem using the group structure. It would surely illustrate the methods we would use later in analysing the higher group size models. I've done it on paper and I don't think I've made any mistakes. However I don't obtain the correct answers. I suspect it's because of the following - $\sigma_1 = 1/(tS_1^1) - S_1^0/S_1^1$ which does not expand as a power series in t. Perhaps we just cannot plug this solution into the functional equations. Therefore one would have to resort to using the σ_2 solution without being able to use the group structure at all. I can certainly solve the problem in the above manner and use the series generated for each part of the solution to test whether the second solution really fails, and if it does test if it fails because of the σ_1 behavior.

Chapter 6

Solving case 3.3 - supersymmetric models of order 4

When we consider the supersymmetric cases the x-transformations are given by

$$\bullet : x_1 \mapsto \overline{x_1}, \quad * : x_2 \mapsto \overline{x_2}$$

In each case we have

$$K^{\bullet} = K^{*} = K, \quad K_2^{\bullet} = K_2, \quad K_1^{*} = K_1$$

while the others change in some way. The solutions eliminating the kernel also have a nice form

$$\sigma_i = \frac{1 - S_i^0 - \sqrt{(1 - S_i^0)^2 - 4t^2 (S_i^1)^2}}{2tS_i^1}, \quad \overline{\sigma_i} = \frac{1 - S_i^0 + \sqrt{(1 - S_i^0)^2 - 4t^2 (S_i^1)^2}}{2tS_i^1}$$

because we have

$$\overline{\sigma}_i = \frac{S_i^1}{S_i^{-1}} conj(\sigma_i) = conj(\sigma_i)$$

We can take the master equation and apply $*$ to it

$$-KQ + K_1 Q_1^0 + K_2 Q_2^0 - K_{1,2} Q_{1,2}^{0,0} + \alpha = 0$$
$$-KQ^* + K_1 Q_1^{0,*} + K_2^* Q_2^0 - K_{1,2}^* Q_{1,2}^{0,0} + \alpha = 0$$

From here on we will input $x_1 = \sigma_1$ to rid ourselves of the kernel K. There will be an overloading of notation

$$K_1 Q_1^0 + K_2 Q_2^0 - K_{1,2} Q_{1,2}^{0,0} + \alpha = 0$$
$$K_1 Q_1^{0,*} + K_2^* Q_2^0 - K_{1,2}^* Q_{1,2}^{0,0} + \alpha = 0$$

We can multiply both equation by something to clear Q_2^0 because it involves σ_1 while the other term involves only x_2. We have

$$K_1 K_2^* Q_1^0 + K_2^* K_2 Q_2^0 - K_2^* K_{1,2} Q_{1,2}^{0,0} + \alpha K_2^* = 0$$
$$K_1 K_2 Q_1^{0,*} + K_2 K_2^* Q_2^0 - K_2 K_{1,2}^* Q_{1,2}^{0,0} + \alpha K_2 = 0$$

We subtract these two equations to obtain

$$K_1 (K_2^* Q_1 - K_2 Q_1^*) - (K_2^* K_{1,2} - K_2 K_{1,2}^*) Q_{1,2} + \alpha (K_2^* - K_2) = 0$$
$$K_2^* Q_1 - K_2 Q_1^* - \frac{K_2^* K_{1,2} - K_2 K_{1,2}^*}{K_1} Q_{1,2} + \alpha \frac{K_2^* - K_2}{K_1} = 0$$

The differences in solutions will occur from here on out.

6.1 Cartesian model: steps N, S, E, W

I haven't written it all up yet but we have the solution

$$Q = \frac{(1 - b_1 b_2 G)(\alpha - \overline{b}_1 K_1 \widehat{K}_1^\bullet - \overline{b}_2 K_2 \widehat{K}_2^* \widehat{Q})}{K - \overline{c}_1 K_1 K_1^\bullet - \overline{c}_2 K_2 K_2^*}$$

6.2 Diagonal model: steps NE, NW, SE, SW

In this case the roots of the kernel become

$$\sigma_i = \frac{1-\sqrt{1-4t^2(S_i^1)^2}}{2tS_i^1}, \quad \overline{\sigma_i} = \frac{1+\sqrt{1-4t^2(S_i^1)^2}}{2tS_i^1}$$

with the extra property that

$$\sigma_1 + \overline{\sigma_1} = \frac{1}{tS_1^{-1}}.$$

First we have

$$\begin{aligned} K_2^* - K_2 &= t(\overline{x_2} - x_2)S_2^{-1}(\sigma_1) \\ &= t(\overline{x_2} - x_2)(\overline{\sigma_1} + \sigma_1) \\ &= \frac{\overline{x_2} - x_2}{\overline{x_2} + x_2} \end{aligned}$$

We have

$$\begin{aligned} K_2^* K_{1,2} - K_2 K_{1,2}^* &= (c_2 - tx_2S_2^{-1})(c_1c_2 - t\overline{\sigma_1}\overline{x_2}) - (c_2 - t\overline{x_2}S_2^{-1})(c_1c_2 - t\overline{\sigma_1}x_2) \\ &= c_1c_2t(\overline{x_2} - x_2)S_2^{-1} - c_2t\overline{\sigma_1}(\overline{x_2} - x_2) \\ &= c_2t(\overline{x_2} - x_2)(c_1(\sigma_1 + \overline{\sigma_1}) - \overline{\sigma_1}) \end{aligned}$$

If we let $v = \overline{x_2} + x_2$ we find

$$\sigma_1 = \frac{1-\sqrt{1-4t^2v^2}}{2tv}, \quad \overline{\sigma_1} = \frac{1+\sqrt{1-4t^2v^2}}{2tv}$$

We have

$$\begin{aligned} (c_1(\sigma_1 + \overline{\sigma_1}) - \overline{\sigma_1}) &= \frac{c1}{tv} - \frac{1+\sqrt{1-4t^2v^2}}{2tv} \\ &= \frac{1}{\overline{x_2} + x_2}\left(-\overline{a_1} + \frac{1-\sqrt{1-4t^2v^2}}{2}\right) \end{aligned}$$

Also

$$\begin{aligned}
K_1 &= c_1 - tv\overline{\sigma_1} \\
&= c_1 - tv\frac{1+\sqrt{1-4t^2v^2}}{2tv} \\
&= 1 - \overline{a_1} - \frac{1+\sqrt{1-4t^2v^2}}{2} \\
&= -\overline{a_1} + \frac{1-\sqrt{1-4t^2v^2}}{2}
\end{aligned}$$

Thus we finally have

$$\begin{aligned}
\frac{1}{K_1}(K_2^*K_{1,2} - K_2K_{1,2}^*) &= -\frac{c_2t(\overline{x_2}-x_2)(c_1(\sigma_1+\overline{\sigma_1})-\overline{\sigma_1})}{c_1 - t\overline{\sigma_1}v} \\
&= -\frac{c_2t(\overline{x_2}-x_2)(-1)}{tv} \\
&= \frac{c_2(\overline{x_2}-x_2)}{\overline{x_2}+x_2}
\end{aligned}$$

Thus the equation will now read as

$$K_2^*Q_1 - K_2Q_1^* - \frac{c_2(\overline{x_2}-x_2)}{\overline{x_2}+x_2}Q_{1,2} + \alpha\frac{\overline{x_2}-x_2}{(\overline{x_2}+x_2)K_1} = 0$$

We just need to take care of the remaining terms. We'll do this by multiplying by $\overline{x_2}+x_2$. We have

$$\begin{aligned}
R^* &= (\overline{x_2}+x_2)K_2^* = c_2(\overline{x_2}+x_2) - x_2 = c_2\overline{x_2} - \overline{a_2}x_2 \\
R &= (\overline{x_2}+x_2)K_2 = c_2(\overline{x_2}+x_2) - \overline{x_2} = c_2x_2 - \overline{a_2x_2}
\end{aligned}$$

We have a couple of relations here

$$\begin{aligned}
R^* - R &= v(K_2^* - K_2) = \overline{x_2} - x_2 \\
\overline{x_2}R^* - x_2R &= \overline{x_2}c_2v - 1 - x_2c_2v + 1 \\
&= c_2v(\overline{x_2} - x_2)
\end{aligned}$$

The equation becomes

$$(c_2\overline{x_2} - \overline{a_2}x_2)Q_1^0(x_2) + (\overline{a_2x_2} - c_2x_2)Q_1^0(\overline{x_2}) - c_2(\overline{x_2} - x_2)Q_{1,2}^{0,0} + (\overline{x_2} - x_2)\overline{K}_1\alpha = 0$$

and it's ready for extraction. Let $\delta_0 = \overline{a_1}(\overline{x_2} - x_2)\overline{K}_1$. We re-write the equation as

$$R^*Q_1^0 - RQ_1^{0,*} - c_2(\overline{x_2} - x_2)Q_{1,2}^{0,0} + \overline{a_2}\delta_0 = 0$$

We'll use the master equation

$$(1 - tS)Q - (c_1 - t\overline{x_1}S_1^{-1})Q_1^0 - (c_2 - t\overline{x_2}S_2^{-1})Q_2^0 + (c_1c_2 - t\overline{x_1x_2}S_{1,2}^{-1,-1})Q_{1,2}^{0,0} = \overline{a_1a_2}$$

to further our work here. For diagonal steps it is

$$(1 - tS)Q - (c_1 - tv\overline{x_1})Q_1^0 - (c_2 - t\overline{x_2}S_2^{-1})Q_2^0 + (c_1c_2 - t\overline{x_1x_2})Q_{1,2}^{0,0} = \overline{a_1a_2}$$

extracting the $[x_1^0]$ coefficient yields

$$Q_1^0 = \overline{a_2} + c_2Q_{1,2}^{0,0} + a_1tvQ_1^1$$

We can extract higher coefficients too. Let $k \geq 1$. Then the $[x_2^k]$ coefficient is

$$\begin{aligned} 0 &= Q_1^k - tS_1^{-1}Q_1^{k+1} - tS_1^0Q_1^k - tS_1^1Q_1^{k-1} - c_2Q_{1,2}^{k,0} + t\overline{x_2}(S_{1,2}^{-1,-1}Q_{1,2}^{k+1,0} + S_{1,2}^{0,-1}Q_{1,2}^{k,0} + S_{1,2}^{1,-1}Q_{1,2}^{k-1,0}) \\ &= Q_1^k - tvQ_1^{k+1} - tvQ_1^{k-1} - c_2Q_{1,2}^{k,0} + t\overline{x_2}(Q_{1,2}^{k+1,0} + Q_{1,2}^{k-1,0}) \end{aligned}$$

We can re-write the last one in a form more convenient for the next few steps

$$Q_1^k = tv(Q_1^{k+1} + Q_1^{k-1}) + c_2Q_{1,2}^{k,0} - t\overline{x_2}(Q_{1,2}^{k+1,0} + Q_{1,2}^{k-1,0})$$

We'll use induction to prove a nice formula but for that we'll first need to establish another base case because of this splitting of cases for the coefficient extraction.

We have

$$Q_1^0 = \overline{a_2} + c_2 Q_{1,2}^{0,0} + a_1 tv Q_1^1$$
$$Q_1^{0,*} = \overline{a_2} + c_2 Q_{1,2}^{0,0} + a_1 tv Q_1^{1,*}$$

Combining them with equation (6.2) gives

$$\begin{aligned}
0 &= R^* Q_1^0 - R Q_1^{0,*} - c_2(\overline{x_2} - x_2) Q_{1,2}^{0,0} + \overline{a_2}\delta_0 \\
&= R^*(\overline{a_2} + c_2 Q_{1,2}^{0,0} + a_1 tv Q_1^1) - R(\overline{a_2} + c_2 Q_{1,2}^{0,0} + a_1 tv Q_1^{1,*}) - c_2(\overline{x_2} - x_2) Q_{1,2}^{0,0} + \overline{a_2}\delta_0 \\
&= a_1 tv(R^* Q_1^1 - R Q_1^{1,*}) + (R^* - R)(\overline{a_2} + c_2 Q_{1,2}^{0,0}) - c_2(\overline{x_2} - x_2) Q_{1,2}^{0,0} + \overline{a_2}\delta_0 \\
&= a_1 tv(R^* Q_1^1 - R Q_1^{1,*}) + \overline{a_2}(\overline{x_2} - x_2) + \overline{a_2}\delta_0 \\
&= a_1 tv(R^* Q_1^1 - R Q_1^{1,*}) + \overline{a_2}((\overline{x_2} - x_2) + \delta_0)
\end{aligned}$$

So we can write the equation

$$R^* Q_1^1 - R Q_1^{1,*} + \overline{a_2}\delta_1 = 0$$

where

$$\delta_1 = \frac{(\overline{x_2} - x_2) + \delta_0}{a_1 tv}$$

This equation together with equation (6.2) form the base cases for the following lemma

Lemma 1. Let $m \geq 0$ be an integer. Then

$$R^* Q_1^m - R Q_1^{m,*} - c_2(\overline{x_2} - x_2) Q_{1,2}^{m,0} + \overline{a_2}\delta_m = 0 \tag{6.1}$$

where δ_m is not a function of a_2.

Proof. We'll show this by induction. We have the base cases $m = 0$ and $m = 1$. The latter one fits the pattern because $Q_{1,2}^{1,0} = 0$ since we cannot have any walks end on coordinates $(1,0)$. Now we assume that the formula is true for all cases up to

and including m. Then we use formula (6.2)

$$\begin{aligned}
&R^*Q_1^m - RQ_1^{m,*} \\
=&tv(R^*Q_1^{m+1} - RQ_1^{m+1,*}) + tv(R^*Q_1^{m-1} - RQ_1^{m-1,*}) \\
&+ c_2(R^* - R)Q_{1,2}^{m,0} - t(\overline{x_2}R^* - x_2R)(Q_{1,2}^{m+1,0} + Q_{1,2}^{m-1,0}) \\
=&tv(R^*Q_1^{m+1} - RQ_1^{m+1,*}) + tv(c_2(\overline{x_2} - x_2)Q_{1,2}^{m-1,0} - \overline{a_2}\delta_{m-1}) \\
&+ c_2(\overline{x_2} - x_2)Q_{1,2}^{m,0} - c_2tv(\overline{x_2} - x_2)(Q_{1,2}^{m+1,0} + Q_{1,2}^{m-1,0}) \\
=&tv(R^*Q_1^{m+1} - RQ_1^{m+1,*}) + c_2(\overline{x_2} - x_2)Q_{1,2}^{m,0} - c_2tv(\overline{x_2} - x_2)Q_{1,2}^{m+1,0}) - \overline{a_2}tv\delta_{m-1})
\end{aligned}$$

Combining this with equation (6.1) leads to

$$\begin{aligned}
0 &= tv(R^*Q_1^{m+1} - RQ_1^{m+1,*}) - c_2tv(\overline{x_2} - x_2)Q_{1,2}^{m+1,0} - \overline{a_2}tv\delta_{m-1} + \overline{a_2}\delta_m \\
0 &= R^*Q_1^{m+1} - RQ_1^{m+1,*} - c_2(\overline{x_2} - x_2)Q_{1,2}^{m+1,0} + \overline{a_2}\delta_{m+1}
\end{aligned}$$

where

$$\delta_{m+1} = \frac{\delta_m}{tv} - \delta_{m-1}$$

This function δ_{m+1} is also not dependent on a_2 and this completes the proof! □

There is an equivalent statement for Q_2^n. We can expand R^* and R to a form we can use

$$(c_2\overline{x_2} - \overline{a_2}x_2)Q_1^m(x_2) - (c_2x_2 - \overline{a_2x_2})Q_1^m(\overline{x_2}) - c_2(\overline{x_2} - x_2)Q_{1,2}^{m,0} + \overline{a_2}\delta_m = 0$$

We start by extracting the $[x_2^1]$ coefficient

$$\begin{aligned}
&c_2Q_{1,2}^{m,2} - \overline{a_2}Q_{1,2}^{m,0} - c_2Q_{1,2}^{m,0} + c_2Q_{1,2}^{m,0} + [x_2^1]\overline{a_2}\delta_m = 0 \\
&c_2Q_{1,2}^{m,2} - \overline{a_2}Q_{1,2}^{m,0} + [x_2^1]\overline{a_2}\delta_m = 0
\end{aligned}$$

We can also extract the $[x_2^{n+1}]$ coefficient for $n \geq 2$

$$c_2Q_{1,2}^{m,n+2} - \overline{a_2}Q_{1,2}^{m,n} + [x_2^{n+1}]\overline{a_2}\delta_m = 0$$

This formula also contains the $n = 0$ case so we treat them together. We can set $a_2 = 1$ to get

$$-\widehat{Q}^{m,n}_{1,2} + [x_2^{n+1}]\delta_m = 0$$

Substituting this back into the above equation gives

$$b_2 Q^{m,n+2}_{1,2} = Q^{m,n}_{1,2} - \widehat{Q}^{m,n}_{1,2}$$

An equivalent formula can be derived if the process is followed through the use of x_1 instead of x_2 in the analysis.

Corollary 2. For integers $m, n \geq 0$ we have the following relations

$$b_1 Q^{m+2,n}_{1,2} = Q^{m,n}_{1,2} - \widehat{Q}^{m,n}_{1,2}$$
$$b_2 Q^{m,n+2}_{1,2} = Q^{m,n}_{1,2} - \widehat{Q}^{m,n}_{1,2}$$

We can sum up those equations to obtain functional equations

Corollary 3. For integers $m \geq 0$ we have the following relations

$$Q^m_1 - \widehat{Q}^m_1 = b_1 Q^{m+2}_1$$
$$\widehat{Q}^m_1 - \widehat{\widehat{Q}}^m_1 = b_1 \widehat{Q}^{m+2}_1$$
$$x_1^2(Q^n_2 - \widehat{Q}^n_2) = b_1(Q^n_2 - Q^{1,n}_{1,2}x_1 - Q^{0,n}_{1,2})$$
$$x_1^2(\widehat{Q}^n_2 - \widehat{\widehat{Q}}^n_2) = b_1(\widehat{Q}^n_2 - \widehat{Q}^{1,n}_{1,2}x_1 - \widehat{Q}^{0,n}_{1,2})$$
$$x_1^2(Q - \widehat{Q}) = b_1(Q - x_1 Q^1_1 - Q^0_1)$$
$$x_1^2(\widehat{Q} - \widehat{\widehat{Q}}) = b_1(\widehat{Q} - x_1 \widehat{Q}^1_1 - \widehat{Q}^0_1)$$

Corollary 4. For integers $n \geq 0$ we have the following relations

$$\begin{aligned}
Q_2^n - \widehat{Q}_2^n &= b_2 Q_2^{n+2} \\
\widehat{Q}_2^n - \widehat{\widehat{Q}}_2^n &= b_2 \widehat{Q}_2^{n+2} \\
x_2^2(Q_1^m - \widehat{Q}_1^m) &= b_2(Q_1^m - Q_{1,2}^{m,1} x_1 - Q_{1,2}^{m,0}) \\
x_2^2(\widehat{Q}_1^m - \widehat{\widehat{Q}}_1^m) &= b_2(\widehat{Q}_1^m - \widehat{Q}_{1,2}^{m,1} x_1 - \widehat{Q}_{1,2}^{m,0}) \\
x_2^2(Q - \widehat{Q}) &= b_2(Q - x_2 Q_2^1 - Q_2^0) \\
x_2^2(\widehat{Q} - \widehat{\widehat{Q}}) &= b_2(\widehat{Q} - x_2 \widehat{Q}_2^1 - \widehat{Q}_2^0)
\end{aligned}$$

Every other equation is obtained from the previous one by either setting $a_1 = 1$ or $a_2 = 1$. Here we can hope to set some telescoping sum type of thing in order to say more about $Q_{1,2}^{0,0}$

Now we can move onto the main result about the diagonal walks and it is that the function for labelled walks ending at $(0,0)$ is connected to the functions for semi-labelled and unlabelled walks endind at $(0,0)$. We can use the master equation to extract the $[x_1^0 x_2^0]$ and $[x_1^1 x_2^1]$ coefficients yielding the following two relations

$$\begin{aligned}
Q_{1,2}^{0,0} &= 1 + t a_1 a_2 Q_{1,2}^{1,1} \\
Q_{1,2}^{1,1} &= t(Q_{1,2}^{0,0} + Q_{1,2}^{2,0} + Q_{1,2}^{0,2} + Q_{1,2}^{2,2})
\end{aligned}$$

We also have

$$\begin{aligned}
b_1 b_2 Q_{1,2}^{2,2} &= b_1(Q_{1,2}^{2,0} - \widehat{Q}_{1,2}^{2,0}) \\
&= b_1 Q_{1,2}^{2,0} - b_1 \widehat{Q}_{1,2}^{2,0} \\
&= Q_{1,2}^{0,0} - \widehat{Q}_{1,2}^{0,0} - \widehat{Q}_{1,2}^{0,0} + \widehat{\widehat{Q}}_{1,2}^{0,0}
\end{aligned}$$

together with

$$\begin{aligned}
b_1 b_2 Q_{1,2}^{2,0} &= b_2 Q_{1,2}^{0,0} - b_2 \widehat{Q}_{1,2}^{0,0} \\
b_1 b_2 Q_{1,2}^{0,2} &= b_1 Q_{1,2}^{0,0} - b_1 \widehat{Q}_{1,2}^{0,0}
\end{aligned}$$

We we have

$$\begin{aligned}
b_1b_2Q^{1,1}_{1,2} &= t(b_1b_2Q^{0,0}_{1,2} + b_1b_2Q^{2,0}_{1,2} + b_1b_2Q^{0,2}_{1,2} + b_1b_2Q^{2,2}_{1,2}) \\
&= t(b_1b_2Q^{0,0}_{1,2} + b_2Q^{0,0}_{1,2} - b_2\widehat{Q}^{0,0}_{1,2} + b_1Q^{0,0}_{1,2} - b_1\widehat{Q}^{0,0}_{1,2} + Q^{0,0}_{1,2} - \widehat{Q}^{0,0}_{1,2} - \widehat{Q}^{0,0}_{1,2} + \widehat{\widehat{Q}}^{0,0}_{1,2}) \\
&= t((b_1b_2 + b_1 + b_2 + 1)Q^{0,0}_{1,2} - (b_2+1)\widehat{Q}^{0,0}_{1,2} - (b_1+1)\widehat{Q}^{0,0}_{1,2} + \widehat{\widehat{Q}}^{0,0}_{1,2}) \\
&= t(a_1a_2Q^{0,0}_{1,2} - a_2\widehat{Q}^{0,0}_{1,2} - a_1\widehat{Q}^{0,0}_{1,2} + \widehat{\widehat{Q}}^{0,0}_{1,2})
\end{aligned}$$

We can finally write

$$\begin{aligned}
b_1b_2Q^{0,0}_{1,2} &= b_1b_2 + ta_1a_2b_1b_2Q^{1,1}_{1,2} \\
&= b_1b_2 + t^2a_1a_2(a_1a_2Q^{0,0}_{1,2} - a_2\widehat{Q}^{0,0}_{1,2} - a_1\widehat{Q}^{0,0}_{1,2} + \widehat{\widehat{Q}}^{0,0}_{1,2})
\end{aligned}$$

Let $G(b_1,b_2)$ stand for $Q(0,0,b_1,b_2)$ and G_i has $b_i = 0$. Then we can write it all as

$$b_1b_2G = b_1b_2 + t^2a_1a_2(a_1a_2G - a_2G_1 - a_1G_2 + G_{1,2})$$

or written as a linear combination

$$(b_1b_2 - t^2a_1^2a_2^2)G + t^2a_1a_2^2G_1 + t^2a_1^2a_2G_2 - t^2a_1a_2G_{1,2} = b_1b_2$$

We can divide out by a_1a_2 and obtain the equation

$$(c_1c_2 - a_1a_2t^2)G + a_2t^2G_1 + a_1t^2G_2 - t^2G_{1,2} = c_1c_2$$

from which we can read off

$$f(a_1,a_2,t) = c_1c_2 - a_1a_2t^2$$

and

$$G = \frac{(c_1c_2 - t^2(a_2G_1 + a_1G_2 - G_{1,2})}{f(a_1,a_2,t)}$$

together with $B(fG) = c_1c_2$ Also, we can solve for Q^j_i and friends, in terms of various $[b^l_k]$ extractions, so overall it should lead to a relationship of the form

$B(FQ) = 0$.

6.3 The almost-full model: missing steps E and W

6.4 The full model: all steps available

6.5 Conjectured relationships depending on $S_{1,2}^{-1,-1}$

As we have seen whenever $S_{1,2}^{-1,-1} = 0$, α and G tend to go together in the equations, so in the end we have a solution that depends on $\beta(G)$ multiplied by G_1, G_2 and/or $G_{1,2}$. In the end this leads to a rational function solution for G in terms of G_1, G_2 and $G_{1,2}$. It's then easy to conjecture that this happens whenever we can have such an algebraic relationship. The other part of the conjecture is that we'd have a linear relationship between G, G_1, G_2 and $G_{1,2}$ when $S_{1,2}^{-1,-1} \neq 0$.

Chapter 7

A little aside

If we could prove the validity of one of the functional equations in the corollaries (5th row in each corollary) we would then be able to extract coefficients and obtain the formulas in the first corollary. However, since we have a direct proof already, this section is kind of irrelevant now. We'll just keep it for now because it has some interesting math. The steps from now on are almost exactly the same as those that you'd have to do to prove the Dyck path formula. We'll prove the last one involving a b_1 coefficient. First we have

$$\begin{aligned} K &= 1 - tS = 1 - t(x_1 + \overline{x_1})(x_2 + \overline{x_2}) \\ K_1 &= c_1 - t\overline{x_1}S_1^{-1} = c_1 + \widehat{K}_1 \\ K_2 &= c_2 - t\overline{x_2}S_2^{-1} \\ K_{1,2} &= c_1c_2 - t\overline{x_1x_2} = c_1c_2 + \widehat{K}_{1,2} \end{aligned}$$

We also need the coefficient extractions

$$\begin{aligned} [x_1^0]: \quad & Q_1^0 - c_2Q_{1,2}^{0,0} - ta_1S_1^{-1}Q_1^1 = \overline{a_2} \\ [x_1^0]: & Q_1^1 - tS_1^{-1}Q_1^2 - tS_1^{-1}Q_1^0 + t\overline{x_2}Q_{1,2}^{2,0} + t\overline{x_2}Q_{1,2}^{0,0} = 0 \\ & -t\overline{x_1}S_1^{-1}Q_1^2 + t\overline{x_1x_2}Q_{1,2}^{2,0} = t\overline{x_1}S_1^{-1}Q_1^0 - t\overline{x_1x_2}Q_{1,2}^{0,0} - \overline{x_1}Q_1^1 \end{aligned}$$

which we'll use soon. We first start with

$$\begin{aligned}
KQ &= K_1 Q_1^0 + K_2 Q_2^0 - K_{1,2} Q_{1,2}^{0,0} + \overline{a_1 a_2} \\
&= (c_1 + \widehat{K}_1) Q_1^0 + K_2 Q_2^0 - (c_1 c_2 + \widehat{K}_{1,2}) Q_{1,2}^{0,0} + \overline{a_1 a_2} \\
K\widehat{Q} &= \widehat{K}_1 \widehat{Q}_1^0 + K_2 \widehat{Q}_2^0 - \widehat{K}_{1,2} \widehat{Q}_{1,2}^{0,0} + \overline{a_2}
\end{aligned}$$

Then we have

$$\begin{aligned}
KQ - K\widehat{Q} &= \widehat{K}_1 (Q_1^0 - \widehat{Q}_1^0) + \widehat{K}_2 (Q_2^0 - \widehat{Q}_2^0) - \widehat{K}_{1,2} (Q_{1,2}^{0,0} - \widehat{Q}_{1,2}^{0,0}) + (c_1 Q_1^0 - c_1 c_2 Q_{1,2}^{0,0} + \overline{a_1 a_2} - \overline{a_2}) \\
&= \widehat{K}_1 (Q_1^0 - \widehat{Q}_1^0) + \widehat{K}_2 (Q_2^0 - \widehat{Q}_2^0) - \widehat{K}_{1,2} (Q_{1,2}^{0,0} - \widehat{Q}_{1,2}^{0,0}) + t b_1 S_1^{-1} Q_1^1
\end{aligned}$$

Also

$$\begin{aligned}
\widehat{K}_1 (Q_1^0 - \widehat{Q}_1^0) - \widehat{K}_{1,2} (Q_{1,2}^{0,0} - \widehat{Q}_{1,2}^{0,0}) =& \widehat{K}_1 b_1 Q_1^2 - \widehat{K}_{1,2} b_1 Q_{1,2}^{2,0} \\
=& b_1 (\widehat{K}_1 Q_1^2 - \widehat{K}_{1,2} Q_{1,2}^{0,0} \\
=& b_1 (-t\overline{x_1} S_1^{-1} Q_1^2 + t\overline{x_1 x_2} Q_{1,2}^{2,0}) \\
=& b_1 (t\overline{x_1} S_1^{-1} Q_1^0 - t\overline{x_1 x_2} Q_{1,2}^{0,0} - \overline{x_1} Q_1^1)
\end{aligned}$$

Next

$$\begin{aligned}
K_2 (Q_2^0 - \widehat{Q}_2^0) &= (c_2 - t\overline{x_1}^2 S_2^{-1}) \overline{x_2} b_1 (Q_2^0 - Q_{1,2}^{0,0}) \\
&= -t\overline{x_1}^2 \overline{x_2} b_1 S_2^{-1} (Q_2^0 - Q_{1,2}^{0,0}) + c_2 b_1 \overline{x_1}^2 (Q_2^0 - Q_{1,2}^{0,0})
\end{aligned}$$

We can finally combine these expressions into

$$x_1^2 (KQ - K\widehat{Q}) = b_1 x_1^2 (t S_1^{-1} Q_1^1 + t\overline{x_1} S_1^{-1} Q_1^0 - t\overline{x_1 x_2} Q_{1,2}^{0,0} - \overline{x_1} Q_1^1 - t\overline{x_1}^2 \overline{x_2} S_2^{-1} (Q_2^0 - Q_{1,2}^{0,0}) + c_2 \overline{x_1}^2 (Q_2^0 - Q_{1,2}^{0,0}))$$

We cancel out the b_1 and we have the equation

$$KQ - K(x_1 Q_1^1 + Q_1^0)) = t x_1 S_1^{-1} (x_1 Q_1^1 + Q_1^0) - (t x_1 \overline{x_2} Q_{1,2}^{0,0} + x_1 Q_1^1 + t\overline{x_2} S_2^{-1} (Q_2^0 - Q_{1,2}^{0,0})) - c_2 (Q_2 - Q_{1,2}^{0,0})$$

Note that

$$K + t x_1 S_1^{-1} = 1 - t\overline{x_1} S_1^{-1}$$

Once we expand the KQ term on the LHS we get

$$\begin{aligned}
&c_1 Q_1^0 - t\overline{x_1} S_1^{-1} Q_1^0 + c_2 Q_2^0 - t\overline{x_2} S_1^{-1} Q_2^0 + c_1 c_2 Q_{1,2}^{0,0} + t\overline{x_1 x_2} Q_{1,2}^{0,0} + \overline{a_1 a_2} \\
=&(1 - t\overline{x_1} S_1^{-1})(x_1 Q_1^1 + Q_1^0) - x_1 Q_1^1 + t\overline{x_1 x_2} Q_{1,2}^{0,0} - t\overline{x_2} S_2^{-1} Q_2^0 + c_2 (Q_2^0 - Q_{1,2}^{0,0})
\end{aligned}$$

and after careful book-keeping all the terms cancel out one by one. This completes the long-winded proof of the identity.

Chapter 8

How To Use the template

If I have seen farther it is by standing on the shoulders of Giants.
— Sir Isaac Newton (1855)

This document provides a quick set of instructions for using the `ubcdiss` class to write a dissertation in LaTeX. Unfortunately this document cannot provide an introduction to using LaTeX. The classic reference for learning LaTeX is **?** 's book [**?**]. There are also many freely-available tutorials online; Andy Roberts' online LaTeX tutorials[1] seems to be excellent. The source code for this docment, however, is intended to serve as an example for creating a LaTeX version of your dissertation.

We start by discussing organizational issues, such as splitting your dissertation into multiple files, in Section 8.1. We then cover the ease of managing cross-references in LaTeX in Section 8.2. We cover managing and using bibliographies with BibTeX in Section 8.3. We briefly describe typesetting attractive tables in Section 8.4. We briefly describe including external figures in Section 8.5, and using special characters and symbols in Section 8.6. As it is often useful to track different versions of your dissertation, we discuss revision control further in Section 8.8. We conclude with pointers to additional sources of information in **??**.

[1] http://www.andy-roberts.net/misc/latex/

8.1 Suggested Thesis Organization

The UBC Graduate and Postdoctoral Studies (GPS) specifies a particular arrangement of the components forming a thesis.[2] This template reflects that arrangement.

In terms of writing your thesis, the recommended best practice for organizing large documents in LaTeX is to place each chapter in a separate file. These chapters are then included from the main file through the use of `\include{file}`. A thesis might be described as six files such as `intro.tex`, `relwork.tex`, `model.tex`, `eval.tex`, `discuss.tex`, and `concl.tex`.

We also encourage you to use macros for separating how something will be typeset (e.g., bold, or italics) from the meaning of that something. For example, if you look at `intro.tex`, you will see repeated uses of a macro `\file{}` to indicate file names. The `\file{}` macro is defined in the file `macros.tex`. The consistent use of `\file{}` throughout the text not only indicates that the argument to the macro represents a file (providing meaning or semantics), but also allows easily changing how file names are typeset simply by changing the definition of the `\file{}` macro. `macros.tex` contains other useful macros for properly typesetting things like the proper uses of the latinate *exempli gratiā* and *id est* (i.e., `\eg` and `\ie`), web references with a footnoted URL (`\webref{url}{text}`), as well as definitions specific to this documentation (`\latexpackage{}`).

8.2 Making Cross-References

LaTeX make managing cross-references easy, and the `hyperref` package's `\autoref{}` command[3] makes it easier still.

A thing to be cross-referenced, such as a section, figure, or equation, is *labelled* using a unique, user-provided identifier, defined using the `\label{}` command. The thing is referenced elsewhere using the `\autoref{}` command. For example, this section was defined using:

```
\section{Making Cross-References}
\label{sec:CrossReferences}
```

[2] See http://www.grad.ubc.ca/current-students/dissertation-thesis-preparation/order-components

[3] The `hyperref` package is included by default in this template.

References to this section are made as follows:

```
We then cover the ease of managing cross-references in \LaTeX\
in \autoref{sec:CrossReferences}.
```

`\autoref{}` takes care of determining the *type* of the thing being referenced, so the example above is rendered as

> We then cover the ease of managing cross-references in LaTeX in Section 8.2.

The label is any simple sequence of characters, numbers, digits, and some punctuation marks such as ":" and "–"; there should be no spaces. Try to use a consistent key format: this simplifies remembering how to make references. This document uses a prefix to indicate the type of the thing being referenced, such as `sec` for sections, `fig` for figures, `tbl` for tables, and `eqn` for equations.

For details on defining the text used to describe the type of *thing*, search `diss.tex` and the `hyperref` documentation for `autorefname`.

8.3 Managing Bibliographies with BibTeX

One of the primary benefits of using LaTeX is its companion program, BibTeX, for managing bibliographies and citations. Managing bibliographies has three parts: (i) describing references, (ii) citing references, and (iii) formatting cited references.

8.3.1 Describing References

BibTeX defines a standard format for recording details about a reference. These references are recorded in a file with a `.bib` extension. BibTeX supports a broad range of references, such as books, articles, items in a conference proceedings, chapters, technical reports, manuals, dissertations, and unpublished manuscripts. A reference may include attributes such as the authors, the title, the page numbers, the Document Object Identifier (DOI), or a Unique Resource Locator (URL). A reference can also be augmented with personal attributes, such as a rating, notes, or keywords.

Each reference must be described by a unique *key*.[4] A key is a simple sequence

[4] Note that the citation keys are different from the reference identifiers as described in Section 8.2.

of characters, numbers, digits, and some punctuation marks such as ":" and "–"; there should be no spaces. A consistent key format simiplifies remembering how to make references. For example:

last-name–*year*–*contracted-title*

where *last-name* represents the last name for the first author, and *contracted-title* is some meaningful contraction of the title. Then **?** 's seminal article on aspect-oriented programming [**?**] (published in **?**) might be given the key `kiczales-1997-aop`.

An example of a BibTEX `.bib` file is included as `biblio.bib`. A description of the format a `.bib` file is beyond the scope of this document. We instead encourage you to use one of the several reference managers that support the BibTEX format such as JabRef[5] (multiple platforms) or BibDesk[6] (MacOS X only). These front ends are similar to reference manages such as EndNote or RefWorks.

8.3.2 Citing References

Having described some references, we then need to cite them. We do this using a form of the `\cite` command. For example:

```
\citet{kiczales-1997-aop} present examples of crosscutting
from programs written in several languages.
```

When processed, the `\citet` will cause the paper's authors and a standardized reference to the paper to be inserted in the document, and will also include a formatted citation for the paper in the bibliography. For example:

> **?**] present examples of crosscutting from programs written in several languages.

There are several forms of the `\cite` command (provided by the `natbib` package), as demonstrated in Table 8.1. Note that the form of the citation (numeric or author-year) depends on the bibliography style (described in the next section). The `\citet` variant is used when the author names form an object in the sentence, whereas the `\citep` variant is used for parenthetic references, more like an end-note.

[5] http://jabref.sourceforge.net
[6] http://bibdesk.sourceforge.net

Table 8.1: Available `cite` variants; the exact citation style depends on whether the bibliography style is numeric or author-year.

Variant	Result
`\cite`	Parenthetical citation (e.g., "[**?**]" or "(**?** **?**)")
`\citet`	Textual citation: includes author (e.g., "**?**]" or or "**?** (**?**)")
`\citet*`	Textual citation with unabbreviated author list
`\citealt`	Like `\citet` but without parentheses
`\citep`	Parenthetical citation (e.g., "[**?**]" or "(**?** **?**)")
`\citep*`	Parenthetical citation with unabbreviated author list
`\citealp`	Like `\citep` but without parentheses
`\citeauthor`	Author only (e.g., "**?** ")
`\citeauthor*`	Unabbreviated authors list (e.g., "**?** ")
`\citeyear`	Year of citation (e.g., "**?** ")

8.3.3 Formatting Cited References

BibTEX separates the citing of a reference from how the cited reference is formatted for a bibliography, specified with the `\bibliographystyle` command. There are many varieties, such as `plainnat`, `abbrvnat`, `unsrtnat`, and `vancouver`. This document was formatted with `abbrvnat`. Look through your TEX distribution for `.bst` files. Note that use of some `.bst` files do not emit all the information necessary to properly use `\citet{}`, `\citep{}`, `\citeyear{}`, and `\citeauthor{}`.

There are also packages available to place citations on a per-chapter basis (`bibunits`), as footnotes (`footbib`), and inline (`bibentry`). Those who wish to exert maximum control over their bibliography style should see the amazing `custom-bib` package.

8.4 Typesetting Tables

?] made one grievous mistake in LATEX: his suggested manner for typesetting tables produces typographic abominations. These suggestions have unfortunately been replicated in most LATEX tutorials. These abominations are easily avoided simply by ignoring his examples illustrating the use of horizontal and vertical rules

(specifically the use of `\hline` and `|`) and using the `booktabs` package instead.

The `booktabs` package helps produce tables in the form used by most professionally-edited journals through the use of three new types of dividing lines, or *rules*. Tables 8.1 and 8.2 are two examples of tables typeset with the `booktabs` package. The `booktabs` package provides three new commands for producing rules: `\toprule` for the rule to appear at the top of the table, `\midrule` for the middle rule following the table header, and `\bottomrule` for the bottom-most at the end of the table. These rules differ by their weight (thickness) and the spacing before and after. A table is typeset in the following manner:

```
\begin{table}
\caption{The caption for the table}
\label{tbl:label}
\centering
\begin{tabular}{cc}
\toprule
Header & Elements \\
\midrule
Row 1 & Row 1 \\
Row 2 & Row 2 \\
% ... and on and on ...
Row N & Row N \\
\bottomrule
\end{tabular}
\end{table}
```

See the `booktabs` documentation for advice in dealing with special cases, such as subheading rules, introducing extra space for divisions, and interior rules.

8.5 Figures, Graphics, and Special Characters

Most LaTeX beginners find figures to be one of the more challenging topics. In LaTeX, a figure is a *floating element*, to be placed where it best fits. The user is not expected to concern him/herself with the placement of the figure. The figure should instead be labelled, and where the figure is used, the text should use `\autoref` to reference the figure's label. Figure 8.1 is an example of a figure. A figure is generally included as follows:

```
\begin{figure}
\centering
\includegraphics[width=3in]{file}
```



Figure 8.1: Proof of LaTeX's amazing abilities

```
\caption{A useful caption}
\label{fig:fig-label}   % label should change
\end{figure}
```

There are three items of note:

1. External files are included using the `\includegraphics` command. This command is defined by the `graphicx` package and can often natively import graphics from a variety of formats. The set of formats supported depends on your TeX command processor. Both `pdflatex` and `xelatex`, for example, can import GIF, JPG, and PDF. The plain version of `latex` only supports EPS files.

2. The `\caption` provides a caption to the figure. This caption is normally listed in the List of Figures; you can provide an alternative caption for the LoF by providing an optional argument to the `\caption` like so:

   ```
   \caption[nice shortened caption for LoF]{%
       longer detailed caption used for the figure}
   ```

   GPS generally prefers shortened single-line captions in the LoF: multiple-line captions are a bit unwieldy.

3. The `\label` command provides for associating a unique, user-defined, and descriptive identifier to the figure. The figure can be can be referenced elsewhere in the text with this identifier as described in Section 8.2.

See Keith Reckdahls excellent guide for more details, *Using imported graphics in LaTeX2e*[7].

8.6 Special Characters and Symbols

LaTeX appropriates many common symbols for its own purposes, with some used for commands (i.e., `\{}&%`) and mathematics (i.e., `$^_`), and others are automagi-

[7] http://www.ctan.org/tex-archive/info/epslatex.pdf

Table 8.2: Useful LaTeX symbols

LaTeX	Result	LaTeX	Result
\texttrademark	™	\&	&
\textcopyright	©	\{ \}	{ }
\textregistered	®	\%	%
\textsection	§	\verb!~!	~
\textdagger	†	\$	$
\textdaggerdbl	‡	\^{}	^
\textless	<	_	_
\textgreater	>		

cally transformed into typographically-preferred forms (i.e., – `` '') or to completely different forms (i.e., <>). Table 8.2 presents a list of common symbols and their corresponding LaTeX commands. A much more comprehensive list of symbols and accented characters is available at: http://www.ctan.org/tex-archive/info/symbols/comprehensive/

8.7 Changing Page Widths and Heights

The `ubcdiss` class is based on the standard LaTeX `book` class that selects a line-width to carry approximately 66 characters per line. This character density is claimed to have a pleasing appearance and also supports more rapid reading [**?**]. I would recommend that you not change the line-widths!

8.7.1 The `geometry` Package

Some students are unfortunately saddled with misguided supervisors or committee members whom believe that documents should have the narrowest margins possible. The `geometry` package is helpful in such cases. Using this package is as simple as:

```
\usepackage[margin=1.25in,top=1.25in,bottom=1.25in]{geometry}
```

You should check the package's documentation for more complex uses.

8.7.2 Changing Page Layout Values By Hand

There are some miserable students with requirements for page layouts that vary throughout the document. Unfortunately the `geometry` can only be specified once, in the document's preamble. Such miserable students must set LaTeX's layout parameters by hand:

```
\setlength{\topmargin}{-.75in}
\setlength{\headsep}{0.25in}
\setlength{\headheight}{15pt}
\setlength{\textheight}{9in}
\setlength{\footskip}{0.25in}
\setlength{\footheight}{15pt}

% The *sidemargin values are relative to 1in; so the following
% results in a 0.75 inch margin
\setlength{\oddsidemargin}{-0.25in}
\setlength{\evensidemargin}{-0.25in}
\setlength{\textwidth}{7in}       % 1.1in margins (8.5-2*0.75)
```

These settings necessarily require assuming a particular page height and width; in the above, the setting for `\textwidth` assumes a US Letter with an 8.5" width. The `geometry` package simply uses the page height and other specified values to derive the other layout values. The `layout` package provides a handy `\layout` command to show the current page layout parameters.

8.7.3 Making Temporary Changes to Page Layout

There are occasions where it becomes necessary to make temporary changes to the page width, such as to accomodate a larger formula. The `chngpage` package provides an `adjustwidth` environment that does just this. For example:

```
% Expand left and right margins by 0.75in
\begin{adjustwidth}{-0.75in}{-0.75in}
% Must adjust the perceived column width for LaTeX to get with it.
\addtolength{\columnwidth}{1.5in}
\[ an extra long math formula \]
\end{adjustwidth}
```

8.8 Keeping Track of Versions with Revision Control

Software engineers have used Revision control system (RCS) to track changes to their software systems for decades. These systems record the changes to the source code along with context as to why the change was required. These systems also support examining and reverting to particular revisions from their system's past.

An RCS can be used to keep track of changes to things other than source code, such as your dissertation. For example, it can be useful to know exactly which revision of your dissertation was sent to a particular committee member. Or to recover an accidentally deleted file, or a badly modified image. With a revision control system, you can tag or annotate the revision of your dissertation that was sent to your committee, or when you incorporated changes from your supervisor.

Unfortunately current revision control packages are not yet targetted to non-developers. But the Subversion project's TortoiseSVN[8] has greatly simplified using the Subversion revision control system for Windows users. You should consult your local geek.

A simpler alternative strategy is to create a GoogleMail account and periodically mail yourself zipped copies of your dissertation.

8.9 Recommended Packages

The real strength to LaTeX is found in the myriad of free add-on packages available for handling special formatting requirements. In this section we list some helpful packages.

8.9.1 Typesetting

`enumitem`: Supports pausing and resuming enumerate environments.

`ulem`: Provides two new commands for striking out and crossing out text (`\sout{text}` and `\xout{text}` respectively) The package should likely be used as follows:

```
\usepackage[normalem,normalbf]{ulem}
```

[8] http://tortoisesvn.net/docs/release/TortoiseSVN_en/

to prevent the package from redefining the emphasis and bold fonts.

chngpage: Support changing the page widths on demand.

mhchem: Support for typesetting chemical formulae and reaction equations.

Although not a package, the `latexdiff`[9] command is very useful for creating changebar'd versions of your dissertation.

8.9.2 Figures, Tables, and Document Extracts

pdfpages: Insert pages from other PDF files. Allows referencing the extracted pages in the list of figures, adding labels to reference the page from elsewhere, and add borders to the pages.

subfig: Provides for including subfigures within a figure, and includes being able to separately reference the subfigures. This is a replacement for the older `subfigure` environment.

rotating: Provides two environments, sidewaystable and sidewaysfigure, for typesetting tables and figures in landscape mode.

longtable: Support for long tables that span multiple pages.

tabularx: Provides an enhanced tabular environment with auto-sizing columns.

ragged2e: Provides several new commands for setting ragged text (e.g., forms of centered or flushed text) that can be used in tabular environments and that support hyphenation.

8.9.3 Bibliography Related Packages

bibunits: Support having per-chapter bibliographies.

footbib: Cause cited works to be rendered using footnotes.

[9] http://www.ctan.org/tex-archive/support/latexdiff/

bibentry: Support placing the details of a cited work in-line.

custom-bib: Generate a custom style for your bibliography.

8.10 Moving On

At this point, you should be ready to go. Other handy web resources:

- Common TEX Archive Network (CTAN)[10] is *the* comprehensive archive site for all things related to TEX and LATEX. Should you have some particular requirement, somebody else is almost certainly to have had the same requirement before you, and the solution will be found on CTAN. The links to various packages in this document are all to CTAN.
- An online reference to LATEX commands[11] provides a handy quick-reference to the standard LATEX commands.
- The list of Frequently Asked Questions about TEX and LATEX[12] can save you a huge amount of time in finding solutions to common problems.
- The teTEX documentation guide[13] features a very handy list of the most useful packages for LATEX as found in CTAN.
- The `color`[14] package, part of the graphics bundle, provides handy commands for changing text and background colours. Simply changing text to various levels of grey can have a very dramatic effect.
- If you're really keen, you might want to join the TEX Users Group[15].

[10] http://www.ctan.org
[11] http://www.ctan.org/get/info/latex2e-help-texinfo/latex2e.html
[12] http://www.tex.ac.uk/cgi-bin/texfaq2html?label=interruptlist
[13] http://www.tug.org/tetex/tetex-texmfdist/doc/
[14] http://www.ctan.org/tex-archive/macros/latex/required/graphics/grfguide.pdf
[15] http://www.tug.org

Appendix A

Supporting Materials

This would be any supporting material not central to the dissertation. For example:

- additional details of methodology and/or data;
- diagrams of specialized equipment developed.;
- copies of questionnaires and survey instruments.